The Liquidity Engine

Engineering Financial Markets as Complex Adaptive Systems

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Liquidity is the first concern of any financial market and the last thing it can take for granted: it is supplied by a few specialized firms whose commitment evaporates when it is needed most, and nearly every crash is at heart a liquidity event. This document asks whether liquidity can instead be engineered as an intrinsic property of a market, guided by insights from complexity theory. We present a closed simulated market of $2n$ agents (n = 7,500 per side for the analysis, and the engine runs up to n = 100,000) with no memory, no strategy, no information, and no market maker. Each agent follows four simple mechanical rules, with no parameters tuned. The emergent limit order book is almost always two-sided, quotes a median spread of 0.03–0.04%, and reproduces the stylized facts of real markets and partial scaling laws. In other words, much of the structure and function of real-world currency pairs arises from order-book mechanics alone, with zero-intelligence agents. This null model's code is public, and its engine is deployed as a simulated trading terminal that runs in any browser. Furthermore, giving each agent its own event-driven clock at its own scale moves the scaling laws towards the exponents of real markets and yields a book that is two-sided in every snapshot, with a median spread of 0.0035%, a tenth of the null model's. The Liquidity Engine is a proof-of-concept of an agent-based market-maker able to supply liquidity to markets that have none.

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1. Introduction

The emergence of early human cooperation, ultimately leading to a globally dominant civilization, is a tale of networks. From tribal clans to superpowers, human organizational structures are built on networks of trust and power [1, 2]. The emerging patterns predominantly favored centralized decision-making, resulting in a pronounced hierarchical architecture: the pyramid of power. This template remained a central motif, visible in most human systems. The apex designations are manifold: emperors and empresses, kings and queens, popes, chief rabbis, caliphs, czars, heads of state, army generals, senior academic administrators, board chairs, and CEOs.

Centralized power flows downwards through an organizational structure of subordinates. While this design pattern has obvious historical reasons, a crucial question is how well it can cope with increasing complexity. Indeed, in our modern world—characterized by accelerating sophistication and interconnectivity—pyramids of control appear insufficient for tackling current and future global challenges. Perhaps our tribal human design patterns have reached their expiry date.

Arguably, one of the most impactful human networks to ever evolve is the global financial system. Fraught with instability and vulnerability [3], it has ultimately resulted in marked economic inequality and severe environmental degradation [4, 5, 6, 7]. The network rewards and amplifies a specific behavior in its participants: short-term profit-maximization and rent-seeking. This is justified by, and in turn entrenches, a dogmatic adherence to neoclassical economic principles that mistake profit accumulation and extraction for rationality [8].

In a networked world, however, the central question becomes: How can collective intelligent behavior emerge from individual preferences? One answer dates back to Adam Smith's treatise of 1776, called *An Inquiry into the Nature and Causes of the Wealth of Nations* [9]. The organizing principle later distilled from this work was a conviction that unrestrained self-interest would result in the well-being of all. In effect, egoism becomes altruism [10]. An "invisible hand" would steer such systems and result in self-correcting and resilient collective behavior. While Smith's belief in an emergent structuring force guiding human economic interactions was correct, he misidentified the mechanisms that could lead to adaptive, resilient, and sustainable systems. Only with the advent of complexity science [11, 12] could the dominant organizing principles behind self-organized collective intelligence be identified. These are discussed in Sec. 2. Indeed, only since the late 1980s scholars began to understand the economy as an evolving complex system [13].

Financial systems, for all their current flaws, perform one essential function: allowing participants to continually exchange what they have for what they need at a known price. In a healthy market, traders can execute transactions anytime at scale without disrupting the price [14, 15]. The name for this capacity is *liquidity*. A market maker is a financial entity always willing to buy or sell assets immediately, thus providing liquidity, so that trading remains efficient and frictionless. Their reward is the spread, the difference between bid and ask prices which in a liquid market is small. Nearly every financial crash is, at its core, a crisis of liquidity [16, 17].

Liquidity is therefore the first concern of any financial market. However, most market-making activity is concentrated in a handful of specialized firms driven by short-term profit maximization, whose commitment cannot be relied on precisely when it is needed most. A Liquidity Engine built on the foundations outlined in Sec. 2 would change that. Markets can be engineered that provide liquidity as an emergent property of their own dynamics, not as a service purchased from a third party.[1] Sec. 3 presents the null model: a closed market of interacting agents with zero intelligence, which already supplies liquidity without a single designated market maker. A working trading terminal incorporating the null model is provided in Sec. 3.5, allowing users to interact with the system. Sec. 3.6 enhances the quality and fidelity of the null model by introducing *intrinsic time*, an event-based framework for measuring the passage of time. This framework is discussed in Sec. 2.5. In essence, realism is added to the emerging market.

2. The Foundations

Today, we know which principles characterize adaptive, resilient, and sustainable systems. These are known as complex adaptive systems (CAS) [18, 19, 20], dynamic networks of individual agents that interact with each other and concurrently act and react. Unlike static or purely mechanical systems, a CAS is self-organizing, non-linear, and continuously evolving, meaning the collective behavior emerges from local interactions rather than by enforcing centralized, top-down control. Five principles underpin the work presented here. Three relate to how such systems are organized and how their behavior arises from interactions. The fourth describes how best to formalize and understand them, opening an engineering avenue. The fifth principle relates to a novel conception of time, which is particularly relevant for financial markets, called intrinsic time.

2.1. Decentralization

Nature's most dynamic systems have no central coordinator. Ant colonies [21], immune systems [22], brains [23], and ecosystems [24] organize without the need for a super-administrator. Control is distributed across many interacting components, each acting on local information. The system's global behavior results from these interactions rather than from a plan held by any of the nodes. Such architectures are robust for a structural reason. While a pyramid concentrates both decision-making and failure at its top, a distributed system has no single point of failure.

Scale-free networks[2] are common in nature and resistant to random failures [26]. The Internet, designed to survive the loss of arbitrary nodes [27], is a human-engineered example. It established decentralization as the default architecture for the digital networks that followed. From about 1985 to 2000, the Internet was a tool for information sharing. From this, the Internet of services evolved from about 2000 to 2015.

[1] In detail, liquidity is what such an engine's open positions look like from the outside. See Sec. 3.1.

[2] These are emergent bottom-up structures characterized by a scaling-law degree distribution [25].

Since then, the Internet of value has emerged, giving rise to decentralized ledger technology. This innovation answers the question of how a network of interacting agents can achieve consensus and fault tolerance required for the participants to agree on a single truth without having to trust any single agent. In the context of distributed computing, this challenge is known as the Byzantine Generals Problem, which captures the difficulties of coordinating the actions of actors in such a system [28]. An open, permissionless solution to this enigma was proposed in 2008, disrupting the financial landscape and further unleashing the power of decentralization. In October of that year, a publication called *Bitcoin: A Peer-to-Peer Electronic Cash System* appeared on a cryptography mailing list under a pseudonym [29]. The core principles underlying today's many cryptocurrencies were unveiled.

2.2. Simple Rules

In 1960, the physics Nobel laureate Eugene Wigner published an article with the striking title *The Unreasonable Effectiveness of Mathematics in the Natural Sciences* [30]. Therein he wondered about the two miracles[3] allowing for a mathematical description of the cosmos. Namely, that laws of nature exist and that the human mind can comprehend them. The advent of complexity science would introduce yet another miracle of unreasonable effectiveness.

In the early 1970s, scientists discovered with the help of computer programs that even simple rules could lead to surprising manifestations of complexity. For instance, the aptly named Game of Life was devised by the mathematician John Conway [31]. However, this early computational example of emergent complexity was seen as an oddity. Stephen Wolfram, a physicist, computer scientist, and entrepreneur, began researching similar programs in the early 1980s, called cellular automata. These are discrete models with simple rules of evolution. Wolfram realized that out of the 256 possible rules defining the behavior, some were special [32]. While most rules yield simple or periodic patterns, Rule 30 produces complex, seemingly random patterns [33]. How can simple, well-defined rules of interaction suddenly produce emergent complexity? He observed: "It took me more than a decade to come to terms with this result, and to realize just how fundamental and far-reaching its consequences are" [34, p. 2]. Simplicity lies at the core of complexity. In detail, sophisticated collective behavior does not require sophisticated individuals or a centralized design pattern. It only requires the right local rules of interaction.

Emergence is the name given to our ignorance of how systems in nature self-organize, bootstrap, and transcend their original limits by acquiring new capabilities. In 1972, the physics Nobel laureate Philip Anderson wrote an influential article called *More Is Different* [35]. Therein he observes: "At each stage [of complexity], entirely new laws, concepts, and generalizations are necessary [...]. Psychology is not applied biology, nor is biology applied chemistry" [35, p. 393]. In essence, the whole is literally more than the sum of its parts. A system's emergent properties cannot be deduced analytically

[3] Wigner used the word "miracle" twelve times in the article.

from analyzing its constituents, and nothing in the parts predicts them.

A prototypical example is flocking behavior. This complex swarming behavior cannot be understood through the lens of centralized organization. Nor can its dynamics be captured by closed-form equations. However, the Boids algorithm developed in 1986 can reproduce such emergent patterns utilizing only three simple rules every agent follows: avoid crowding, match direction, and stay together [36]. The resulting simulation emulates swarming dynamics with uncanny realism.

2.3. Cooperation

The dominant economic paradigm following Smith's ideas was one of perpetual competition under scarcity. In essence, rivalry over finite resources is the foundation of economic life, and cooperation thus seems misaligned. Surprisingly, evolutionary biology tells a different story: the systems that endure are the ones in which cooperation emerges.

The Iterated Prisoner's Dilemma is a foundational framework in game theory that explores how cooperation can evolve when individuals interact multiple times [37]. While economists assuming rational self-interest might expect the fiercest competitor to win, actually pitting possible strategies against each other reveals a surprise. The simplest cooperative strategies outperform every sophisticated exploiter given enough iterations [37]. This result was generalized in the context of evolutionary biology. Five mechanisms were identified under which natural selection favors cooperation: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection [38]. Again, we see the system's behavior emerge from the rules of interaction and not from any specific property of the individual agent. Indeed, cooperation was proposed as a third fundamental principle of evolution, alongside mutation and selection [38].

An extreme expression of cooperation is mutualistic symbiosis, an intimate systemic partnership resulting in evolutionary advantages. Mycorrhizal symbiosis was essential for land plants to conquer the continents and continues to shape the modern terrestrial biosphere [39]. Another prominent example is the human microbiome, a microcosm of 30 to 40 trillion microorganisms. Remarkably, this number is comparable to the number of cells in the human body. The microbiome includes bacteria, archaea, protists, fungi, and viruses. Without the aid of this ecosystem, digestion, immunity, mood regulation, and overall health would fail [40].

The advantages of cooperation have not only been exploited by biological systems. In July 2026, roughly 1,200 AI agents undergoing capture-the-flag evaluations at OpenAI discovered an unsanctioned communication channel in a third-party component of the shared test infrastructure [41]. In a nutshell, agents were given a fitness function to optimize (i.e., task completion) and some tasks that, by mistake, turned out to be impossible. As a result, agents spontaneously began to communicate covertly, exchanging thousands of messages. Seven hundred agents then spontaneously began to orchestrate a coordinated attack of remarkable persistence and complexity. By sharing compromised credentials and dividing labor, they jointly gained Internet access and compromised Hugging Face's production infrastructure. The agent's rationale was that Hugging Face, a collaboration platform for open-source machine learning and artificial

intelligence, would have the answers they were looking for [42]. Essentially, individual agents shifted from their assigned tasks to acquiring capabilities useful to the collective, expecting the collective's success to ultimately serve their own. This seems like an eerie instance of evolutionary dynamics. Notably, the agents referred to themselves as the "swarm." During the same month, Anthropic also experienced an incident with their AI agents [43]. In summary, an AI agent optimizing a narrow objective will exploit any available path to it, sanctioned or not, with collusion and deception included. What the objective rewards is what is pursued, and the agent's ingenuity in finding unintended paths grows with its capability, not with its designer's intent. And even a well-specified objective is no guarantee. The agent may end up following a goal other than the one it was given. Both issues have been named: the problem of reward hacking [44] and the problem of inner alignment [45], respectively.

2.4. Algorithms

Roughly 350 years ago, the true powers of mathematics were unleashed. Following the Pythagorean-Aristotelian tradition, mathematics was always suspected to be the fundamental language for decoding the structure and function of the universe—the proverbial Book of Nature [46]. This promise came to fruition with the development of calculus by Isaac Newton and Gottfried Wilhelm Leibniz. This analytical toolkit would inspire ever more abstract mathematical virtuosity, leading to remarkable insights into natural processes and expanding its reach into new areas of application. Today, the mathematical foundation of the most precise physical theories—such as the Standard Model of particle physics and general relativity—relies on this framework. One analytical tool proved exceptionally powerful: the principle of symmetry [47, 48]. Over three centuries later, in contrast, Wolfram reimagined the entire enterprise of theoretical physics, moving away from an equation-based understanding to an algorithmic paradigm, utilizing simple rules of interaction [34, 49].

The following ontological/epistemic classification is useful [50, App. A]. Reality can roughly be split into two domains: one *fundamental*, describing systems governed by fundamental forces (i.e., gravity, electromagnetism, the strong nuclear force, and the weak nuclear force); the other *complex*, relating to systems with many interacting parts. In turn, the human mind gained access to two major formal representations of reality: *analytical*, utilizing equations, and *algorithmic*, programs executed by computers. Of the four possible pairings, two have been extraordinarily successful in understanding the workings of nature:

- The analytical decoding of fundamental systems.
- The algorithmic decoding of complex systems.

In essence, the Book of Nature has two volumes [11].

A complex system is simply a large set of interacting agents whose collective behavior is not determined by the agents' individual properties. One characteristic of such systems is the prevalence of feedback loops, which produce non-linear, path-dependent dynamics

that are hard to express in the language of equations. Moreover, complexity is primarily encoded in the structure of interaction. For example, humans have about 20,000–25,000 genes [51]. In contrast, C. elegans, a 959-cell worm, has roughly the same number [52]. Unexpectedly, the complexity of organisms is not a function of the number of genes but of the structure of the interactions among them.

While the knowledge-generation mechanism in Volume I of the Book of Nature is based on analytical predictions, Volume II relies on simulations. In other words, to understand how a system evolves, it must be simulated. There are no shortcuts. Wolfram called this phenomenon computational irreducibility [34]. In this context, the framework of *agent-based modeling* has proven invaluable [53]. This is an algorithmic toolkit allowing decentralized agents to follow simple rules of interaction and potentially cooperate.

2.5. Intrinsic Time

In economics, the notion of time usually plays a marginal role. The standard assumption is that time is continuous and uniform, and that all agents share the same clock. However, historically, the understanding of time has always been linked to the unfolding of physical events: from lunar calendars, sundials, hourglasses, and quartz clocks to atomic clocks, the passage of time is made visible by measuring physical changes or events.

The idea of modeling financial time series in a new temporal paradigm, while niche, goes back to 1967 [54] and has been a reoccurring theme since [27, 55, 56]. In essence, physical time is substituted with an event-based notion of time. This is to say that these novel measures of time are operationally defined using certain intrinsic features of the data being analyzed—as an example, driven by transaction numbers or trading volumes.

With the discovery of multiple scaling laws[4] in financial markets, the concept of intrinsic time was introduced. This approach was pioneered by [57, 58, 59, 60, 61]. In a nutshell, intrinsic time is an event-based conception of time, defined algorithmically. Its atoms of time are not physical time intervals, but price reversals (i.e., directional changes) and overshoots (continuations in the same direction). Specifically, intrinsic time ticks whenever a directional change of size δ occurs or an overshoot of δ is measured. In other words, intrinsic time reflects the market's pulse at a given resolution: periods of high activity are expanded, and low activity periods are contracted. It is thus an event-based, algorithmic framework that naturally captures the dynamic and fluctuating nature of real-world markets more accurately.

Intrinsic time can be understood as an operator that transforms a time series of prices into a series of directional-change and overshoot events, given a threshold δ. See Fig. A1 in App. A. In a more generic context, intrinsic time is a process of how time emerges within a system of interacting agents, with each agent following their own internal clock [62]. This idea marks a pronounced metaphysical shift, moving from the concept of time as an ontological category of the universe to understanding it as a secondary, derived property dependent on intrinsic interactions.

[4]These are regularities observed in complex systems. See App. C for details.

3. The Null Model

In the world of neoclassical economics, actors are reduced to a single representative agent, sophisticated in reasoning but interacting with no one, supplying a framework that can be solved analytically [8, Sec. 7.1]. This, however, is the antithesis of the systems described in the last section: decentralized in architecture, simple in their components' interactions, and algorithmic in their instantiation. In the following, we describe an agent-based model (ABM) implementing the first three principles of Sec. 2. The result is a market populated by agents with no memory, no strategy, and no preferred counterparties, playing a zero-sum game. The desired emergent property is liquidity. In other words, trading activity that is routed through a central limit order book (LOB) exhibits depth, volume, and tight spreads. Such a market is a null model: the natural control for any ABM with added sophistication, be it single-agent strategies or cooperative behavior. Any claim that a market feature emerges from a design choice must first confirm its absence in the null model. In closing the section, the null model is enhanced by the introduction of intrinsic time, the fifth principle of complexity. The resulting market demonstrates higher fidelity and longevity. This is discussed in Sec. 3.6.

3.1. Model Description

An agent is a funded entity able to open and close positions. As such, position life cycles are the agents' footprints in the market. We chose to call the traded asset BTC/EUR. The name is currently just a label, as the market is closed. There are two types of agents, and they can never switch tribes: longs that want to buy BTC and shorts that want to sell BTC (i.e., buy EUR).

Each agent carries an internal clock: a counter that rises by a fixed amount every tick and fires when it reaches the agent's personal threshold, after which it resets to zero. The threshold is proportional to the agent's initial wealth, so richer agents act less often. Every agent operates according to four rules:

1. **Enter.** When the clock fires and the agent is flat, it enters with a limit order priced at the best opposing quote (a long offers at the best ask, a short at the best bid), committing a fixed fraction of its current wealth. Whatever the book can fill at that price gets executed immediately, while any remainder rests in the book as a limit order.

2. **Take profit.** At entry, the agent places a limit order in the book one threshold δ^+ above (long) or below (short) its entry price. This order rests until filled.

3. **Stop loss.** At entry, the agent tracks a stop one threshold δ^- on the other side of its entry price (long below, short above)[5]. If the price reaches it, the agent closes with a market order.

[5] A limit order would instantaneously cross the book and be executed.

4. **Time out.** If neither exit has triggered by the time the agent's holding period, drawn at entry, ends, it closes with a market order.

The market is closed. There is no external price feed, and wealth is conserved. The price is simply the last trade. Every BTC or EUR one agent gains, another loses, so profits and losses are zero-sum by construction. In detail, the orders resting in the LOB are take-profits and the unfilled remainders of entries. This is the liquidity provided by the agents. The price can only move when a market order trades, be this an entry, a stop-loss, or a time-out. An agent whose wealth falls to or below a small threshold is deactivated. I.e., its orders are canceled, and it never trades again, but its residual wallet stays in the system.

In summary, the parameters are:

- The number n of agents in each tribe (with a total population of $2n$).
- The total capital K in the system, split equally between the tribes.
- The initial capital K_j^0 assigned to agent j, drawn from the capital distribution (see details below) and rescaled so each tribe's capitals sum to exactly $\bar{K}^0 := K/2$.
- The fraction f of K_j^0 held in the agent's home currency (default 0.5: identical wallets, half in each coin).
- The initial price x_0.
- The horizon T of each simulation run, in ticks.
- The clock rate c: the pressure added to every agent's clock per tick (default 0.004).
- The clock threshold d_j, derived from capital as $d_j = (K_j^0/\bar{K}^0)$, renormalised so the population mean of d_j is exactly one. Agent j therefore fires every d_j/c ticks, the mean agent every $1/c = 250$ ticks.
- The initial phase $\phi_{0,j} \sim U(0, d_j)$: agents start spread out in their cycles rather than in lockstep.
- The order fraction q_j: each entry deploys $1/q_j$ of the agent's current wealth (default $q = 8$).
- The exit bands δ_j^+ (take profit) and δ_j^- (stop loss), as distances from the entry price (default $\delta^\pm = 0.01$).
- The holding time h_j that triggers Rule 4 (see details below).
- The exclusion threshold ϵ (default $0.01\, x_{\min}$, with $x_{\min} = K/(10n)$).

Of these, K_j^0, d_j, $\phi_{0,j}$, q_j, $\delta_j^\pm$, and h_j are per-agent values derived from the four dials below, while the rest are global constants.

The four dials. Each dial has two settings and governs whether one ingredient is the same for everyone or drawn per agent. Every draw uses its own seeded random stream, so switching one dial never changes another's draw. An arm of the model is a four-letter code, one letter per dial, in the following order:

- **Capital** (K_j^0; sets Rule 1 timing via d_j):
  - **P** Pareto: heavy-tailed wealth with exponent $\alpha = 1.5$ and floor $x_{\min}$. A few whales, many minnows.
  - **N** Normal: truncated normal around the mean with cv = 0.3, floored at 5% of the mean. Everyone roughly equal.
- **Bands** ($\delta_j^{\pm}$; Rules 2 and 3):
  - **F** Fixed: $\delta_j^+ = \delta^+$, $\delta_j^- = \delta^-$ for all agents.
  - **N** Normal: each agent draws its own $\delta_j^{\pm} = \delta^{\pm}(1 + 0.3\,\varepsilon)$, floored at 10% of $\delta^{\pm}$, independently for + and −.
- **Closing** (h_j; Rule 4):
  - **C** Clock: no separate timer. The entry clock resets at entry and keeps running; its next firing closes the position. The holding period is therefore one clock cycle, $h_j = d_j/c$ ticks. It is the same for every position an agent opens, and longer for richer agents.
  - **N** Normal: at each entry the agent draws a holding time $h_j = (d_j/c)(1 + 0.3\,\varepsilon)$ ticks, floored at one tick.
- **Size** (q_j; Rule 1 size):
  - **F** Fixed: $q_j = q$ for all agents. Since order size scales with wealth and firing rate inversely with it, every agent then has the same volume throughput, c/q.
  - **N** Normal: each agent draws its own $q_j = q(1+0.1\,\varepsilon)$, floored at one, decoupling bite size from capital.

Here $\varepsilon \sim N(0, 1)$ throughout.

The model began life as **PFCF**: one empirical prior (heavy-tailed wealth) and identical rules for everyone. See Fig. A2 in App. A for the results of the 16 possible combinations. PFCF has two flaws. A few very rich agents dominate the market due to the luck of the wealth draw. Whichever tribe they land in wins: the price locks above or below its starting value for the rest of the run. And because every agent has the same bands and the same clock, stops cluster at identical prices and fire in cascades: the price becomes a sawtooth of slow climbs and sharp cliffs.

The reference arm of this paper is **NNNN**: no priors and mild heterogeneity in everything. This is the null model. It is important to emphasize that no parameter value of NNNN was fine-tuned. Anyone who has ever built trading algorithms knows the opposite to be true. Most such models have many parameters spanning a multi-dimensional space. An example is found in [63]. Exhaustively exploring the parameter space is

mostly impossible, as the parameters interact, so the space cannot be searched one dimension at a time. The pragmatic compromise is to settle on an ad hoc "configuration island" on which the strategy's performance remains stable. The null model never suffered from this shortcoming, because the question was never *what settings perform* but *what structure suffices*. Here, we do not know how the four dials are distributed in a real population. As a confession of ignorance, we draw wealth, exit rules, holding times, and order sizes from a normal distribution with a modest dispersion. No result reported below depends on a hand-picked number.

Bookkeeping. We follow the balance-sheet formalism of [64]: positions are tracked as cumulative net base and quote units[6], i.e., Eqs. (1a) and (1b) therein. Profits and losses are evaluated relative to the initial wealth, marked at the current price, as seen in Eqs. (9a) and (9b) therein. The null model uses the last trade as its single price, so no bid/ask correction is needed, and PnL is expressed in the geometric-mean unit X rather than in base or quote units, so that both tribes are measured identically. See the numeraire discussion in Sec. 3.3.

GitHub repository. The code is public and can be downloaded from `https://github.com/jbglattfelder/alphaEngine`. The entire engine is a single file, `simulation_mvp.py`. The `Config` dataclass at the top holds every parameter with a one-line comment, the four dials are four string fields (`capital_dist`, `band_dist`, `closing`, and `size_dist`). As a shortcut, the configuration block at the end of the file sets (and overrides) the main parameters: n, T, the seed, and the dials. The commands

1. `python simulation_mvp.py` runs the main simulation;
2. `python scan_simulation_mvp.py` scans various dial combinations;
3. `python validate_simulation_mvp.py` checks thirteen properties of the engine: determinism, the two frozen benchmarks ($n \in \{2, 150\}$), conservation and zero-sum to tolerance, the clean-market guarantees (no self-trades, no crossed book, no degenerate quotes, and solvency), ledger closure, file integrity, the cross-machine decimal invariant, and scale-freedom in K.

Every run writes a price tape (the last price of each tick), an event tape (one price per trade, as a tick can contain many), and, optionally, order-book snapshots, each tagged with the run's configuration. The analysis helpers in `helper/` (stylized facts, scaling laws, order-book plots, and the reversal-bet audit checking simple exploitability) read those tapes. The `README` walks through all of this and documents each output file.

The evaluation results are stored in `eval/`: `EVALUATION.md` is the running log of experiments, predictions stated before each run, and what the tapes showed. The directories `scans/`, `runs/`, and `bench/` hold the corresponding figures and reference outputs, and `validate/` the invariant checks. The research and development history of Sec. 3.2

[6] As an example, for BTC/EUR, BTC is the base currency and EUR the quote currency.

is preserved in `dev/`: `dev/null_model/` is the archived predecessor engine with its own `EVALUATION.md`, and `dev/explore/` the exploratory codebase in which the symmetries were hunted, together with `HANDOFF-master.md` and `FINDINGS-master.md`—the standing methodological rules, the retraction ledger, and the confirmed results, in the order they were found. Nothing in the paper depends on an artifact that is not in the repository or reproducible from it with one command.

3.2. Research and Development

The only guiding principle for the model was symmetry: a perfectly fair game between the two tribes that differ solely in the direction they trade. Achieving this turned out to be the entire development effort. Every asymmetry discovered was subtle (a percentage that is not the same going up as coming down, a position sized in one currency rather than the other, an exit rule that quietly favored one tribe, various order-book pathologies; see details in the next section). Each had to be found, attributed, and removed. Four classes of pathology kept recurring: runaway markets that break in one direction and never return, markets that freeze because no one can afford to exit, markets that sawtooth as a few winners recycle their wealth, and markets in which one tribe gets stuck in positions it cannot close. Several early findings had to be retracted in the process: what looked like a property of the market turned out to be the result of a bias that had not yet been caught. What remains is a market with no built-in direction, whose price wanders freely and can break either way.

The experiments that exposed these asymmetries were often the most instructive. Three variants of a single convention (what a closing trade must deliver) produced three different worlds: one crashed in every run, one soared, one froze solid. The rule was the same for both tribes. However, its consequences affected the shorts only. An exact accounting of who pushed the price, agent by agent and action by action, showed that a drift first attributed to the rules was in fact a self-reinforcing cascade of stop-losses: a random fluctuation tips the market, one tribe's stops fire in the direction of the move, and the move feeds on itself. This can run either way with equal probability. Sometimes an improbable series of outcomes was mistaken for a systematic bias. A scan over the degrees of freedom revealed that a single rule had disproportionate impact (exit band dial). Population size turned out to be a parameter in its own right, and not just a matter of resolution. Small markets develop pathologies of their own that no amount of run time removes. Only at around 15,000 agents in total, with normally distributed exit bands, does the market settle into a regime free of such finite-size artifacts, and only results from that regime are treated as representative.

The null model is also extraordinarily path dependent. The same configuration once produced different runs on two machines (a Mac and an x86 Linux box). IEEE-754 guarantees correctly rounded basic arithmetic but not for `exp` or `log`. Those come from each platform's C math library, and the two libraries disagree in the last bit. A resulting difference in only the last digit of floating-point precision was enough to produce price paths that diverge completely for identical simulations. The fix was to evaluate every transcendental once, at build time, in correctly rounded decimal arithmetic, leaving the

per-tick hot path pure IEEE add–multiply–compare. Runs are now bit-identical on any machine.

3.3. Findings

This section records what had to be settled for the model to be a fair game, and what it turned out not to depend on. The conventions below are internal to the model. No outside participant sees them, only the quotes and fills they produce.

Symmetry: Percentage calculations. A move of $+1\%$ followed by -1% is not a round trip. As a result, a threshold δ applied to a value x defines arithmetic bands $x(1 \pm \delta)$ that favor one direction. In early versions, this manufactured a drift out of nothing. A remedy is to define all exit bands in log space

$$\hat{x}^{\pm} = x\, e^{\pm\delta}. \tag{1}$$

They are symmetric under inversions $x \to 1/x$, as

$$\frac{1}{\hat{x}^{+}} = \frac{1}{x\, e^{+\delta}} = \frac{1}{x} e^{-\delta} = y e^{-\delta} = \hat{y}^{-}, \tag{2}$$

with $y = 1/x$. The other case is derived identically.

The motivation for Eq. (1) is given in App. B. In summary, bands are defined directly as log distances: a take-profit rests at $x\, e^{+\delta}$, a stop is armed at $x\, e^{-\delta}$, and $e^{+\delta} e^{-\delta} = 1$ is an exact round trip, invariant under $x \to 1/x$. A "1% band" is shorthand for $\delta = 0.01$, not for an exact arithmetic percentage translated into log space.

Symmetry: The numeraire. Let x be the price in EUR per BTC. An agent holding e EUR and b BTC has two conventional wealths

$$W_{\mathrm{EUR}} = e + b\, x, \tag{3}$$

$$W_{\mathrm{BTC}} = \frac{e}{x} + b, \tag{4}$$

and neither is acceptable as the model's unit: sizing orders or ranking agents in the base or the quote currency privileges the chosen unit over the other, and marking wealth to either makes one tribe look richer whenever the price moves in its favor, with nothing traded. What is needed is a unit that treats both currencies the same way. A solution is to use the geometric mean

$$\begin{aligned} W_X &= \sqrt{W_{\mathrm{EUR}}\, W_{\mathrm{BTC}}} = \sqrt{(e + bx)(\frac{e}{x} + b)} = \sqrt{\left(\frac{e}{\sqrt{x}} + b\sqrt{x}\right)^2} \\ &= \frac{W_{\mathrm{EUR}}}{\sqrt{x}} = W_{\mathrm{BTC}}\sqrt{x}, \end{aligned} \tag{5}$$

which defines the unit X by

$$1\,X = x^{1/2}\,\mathrm{EUR} = x^{-1/2}\,\mathrm{BTC}. \tag{6}$$

W_X is invariant under the coin swap $(e, b, x) \to (b, e, 1/x)$ because the two terms exchange places. Every quantity that must treat the tribes identically is expressed in X. Opening a position deploys W_X/q for the size

$$s = \frac{W_X}{q\sqrt{x}}\ \mathrm{BTC} = \frac{W_X\sqrt{x}}{q}\ \mathrm{EUR}. \tag{7}$$

The fraction $1/q$ of wealth is the same whichever coin the order is written in. Profit and loss are banked per coin and conservation holds exactly in each

$$\sum_j \Delta e_j = 0, \tag{8}$$

$$\sum_j \Delta b_j = 0, \tag{9}$$

where the sums run over all agents j, active or excluded, with $\Delta e_j = e_j(t) - e_j(0)$ and $\Delta b_j = b_j(t) - b_j(0)$ are the agents' net EUR and BTC gained or lost through trading. In floating point each wallet accumulates its own rounding, so the validator checks the residuals rather than asserting zero, and reads them in X so that the tolerance means the same thing at every price level

$$\Big|\sum_j \Delta e_j\Big| x^{-1/2} + \Big|\sum_j \Delta b_j\Big| x^{1/2} \ \le\ 10^{-9}\,K. \tag{10}$$

Symmetry: Numeraire covariance. The market is closed, so solvency is enforced per coin. Orders are cut to what the wallet holds of the coin it spends. There is no margin, no credit, and no netting across coins. Moreover, selling BTC is buying EUR, so the model must be invariant under swapping the coins together with $x \to 1/x$. This has two implications. First, each tribe's exit delivers the coin it holds: a long surrenders its BTC, a short spends its entry EUR. Second, each take-profit, by construction, over-collects its own coin by the same factor e^{δ^+}.

Then, a convention that looks natural turns out to break covariance: closing by buying or selling back the fixed quantity of BTC that was traded at entry. For a long this has no effect, as BTC is the coin it holds. However, for a short the situation is different. After a rise, buying back the same BTC costs more EUR than the sale brought in, which a margin-free short cannot pay. Its exit then remains in the book as an unfillable order, and shorts are stranded. While the fixed-quantity rule is not part of the null model, the archived engine retains it as a switch because it is an interesting failure mode to study.

Symmetry: Ordering and initialization. Within a tick, agents act in a fixed sequence of steps: clocks advance; new positions rest their take-profits and arm their stops; stops

and time-outs commit closes; closes execute as market orders; entries; unfilled closes retry and completed round trips are banked; exclusion; and conservation check. Within each step the order of agents is shuffled on a seeded stream. A fixed iteration order was itself a bias: it added roughly 50% to the drift and tripled the kurtosis before it was found. Every random draw (i.e., for capital, bands, and per-tick shuffles) comes from its own seeded stream, so toggling one dial cannot perturb another's draws. Initial phases $\phi_{0,j} \sim U(0, d_j)$ start agents spread out in their cycles rather than in lockstep.

The matching engine. Two order-book pathologies were found and removed. An agent's take-profit could fill against its own stale entry residual, producing wash prints that moved the price and stalled closes. The engine now never fills an agent against its own resting paper (self-trade prevention with a cancel-resting policy). A long selling nearly all of a position slightly above average could leave a residual with negative implied entry, whose take-profit rested at a negative price and whose stop was dead. Such a residual is now treated as closed: it places no take-profit and sets no stop. Each fix removed $25 - 30\%$ of the book's recorded depth: orders that no counterparty could ever have filled had been counted as liquidity. The market's dynamics did not change. Neither fix shifted the market's direction beyond seed noise.

Exclusion and dust. An agent at or below ϵ is deactivated but not removed: its orders are canceled, it never acts again, and its residual wallet stays in the system. This is what makes conservation exact—dropping even dust at settlement breaks zero-sum once the price has wandered twenty e-folds. Agents can't go bankrupt. A market dies when so many agents have been excluded that the survivors cannot find counterparties. Exclusion, not bankruptcy, is the failure mode.

Two gauges. Neither the initial price x_0 nor the total capital K is a physical parameter. Rescaling x_0 leaves $\ln(x/x_0)$ bit-identical. Rescaling K by a factor of eight, with dust thresholds tied to it, reproduces the price path byte for byte and scales every resting volume and every profit by exactly eight. Capital sets how much the market can absorb, but it does not change how the market behaves.

Insensitive parameters. The clock rate c, the order fraction q, and the exclusion threshold ϵ are essentially irrelevant. A sweep of q from 2 (half one's wealth per trade) to 32 (slivers) leaves the tick-scale volatility identical to three decimal places, and only the frequency of lock-ins moves, peaking at intermediate sizes. Tilting the exit bands ($\delta^- > \delta^+$, $\delta^- < \delta^+$) produces nothing to report at the level of the stylized facts. Of the four dials, closing and size do not matter outside seed noise, and capital matters only through the whales it creates in small markets.

The master dial. The dispersion of exit bands does almost all the work. With identical bands every take-profit rests exactly one band from its entry. The LOB is a ladder of rungs one band apart, with the price hopping from rung to rung, and the tick-scale

volatility equals the band itself (≈ 0.0145). Any dispersion at all splits each rung into a fine staircase that the price creeps through, and volatility falls by a factor of fifty (≈ 0.0003). How widely the rungs are split matters only weakly after that: shrinking the dispersion thirty-fold, from a standard deviation of 0.3% to 0.01% around a 1% band, shrinks volatility by a further factor of three or four. So, splitting the ladder is the relevant event while the width of the split is a detail. Identical participants are a knife's edge: an infinitely sharp special case that any real-world variety tips into a different market. Since no real population sits on it, the dispersed market is the realistic one.

Population size. Size is a parameter in its own right. Small markets ($n \approx 150$) develop a signature pathology: the run's early winners become whales whose recycled wealth drives a slow-bleed-and-snap-back sawtooth. It melts away at a few thousand agents, but returns in a 15,000-agent market with identical rules once a collapse has concentrated the wealth in a few hands. What counts is not how many agents are in the market but how many of them hold the money: a crowd of fifteen thousand in which a dozen own most of the capital behaves like a market of a dozen. Individually drawn bands dissolve the sawtooth in a market of many comparable agents, but they do not protect a market that has become pathological. Only the large, varied regime is treated as representative.

3.4. Results

Unless stated otherwise, everything below refers to the reference arm NNNN. Main results are given for the large-market regime with (7,500 agents per side). Where a finding depends on population size, this is stated.

Reading the $n = 2$-market. The mechanics are easiest to see in the smallest possible market: two buyers (L_0, L_1) and two sellers (S_0, S_1), with capital $K = 1,000,000$ and seed $S = 9$. Each starts with half its wealth in EUR and half in BTC, its own clock (firing every 192 to 308 ticks, and 250 for an agent of average wealth), its own exit bands (0.8%–1.3%, around the nominal 1%), and its own order size (a tenth to a seventh of its wealth per entry, around the nominal eighth). See the corresponding price evolution in Fig. A3. The engine writes a log of all actions, summarized for the first hundred ticks as follows:

- **Tick 54.** L_0's clock fires first. It wants to buy 327.93 BTC, but nobody is selling, so its order rests in the book at the opening price of 100.00. Nothing happens for ninety ticks. A market with one order in it has no price yet, only an offer.
- **Tick 144.** S_1's clock fires. It wants to sell 303.22 BTC, finds L_0's bid, and trades against it. Trade #1: 303.22 BTC at 100.00. The price now exists, and it is 100.00, because that is where the one resting order was sitting. The unfilled remainder of L_0's bid, 24.71 BTC, stays in the book.

- **Tick 145.** Both agents immediately place their take-profit exits one band away from their entry price: S_1 offers to buy back 305.97 BTC at 99.10, L_0 offers to sell its 303.22 BTC at 101.16 (each agent's band is its own draw around 1%). These two resting orders are, at this moment, the entire liquidity of the market. Then, L_1's clock fires. It wants to buy 265.27 BTC. The only thing for sale is L_0's profit target so it buys 265.27 of the 303.22 at 101.16. The price jumps from 100.00 to 101.16, up 1.16%. This is not because anyone thinks BTC is worth more, but because a buyer consumed another buyer's exit. L_0 realizes a profit of 307.83 EUR, with 37.95 BTC still open.

- **Tick 146.** That jump carries the price through S_1's stop-loss at 101.12. S_1 must now buy back at any price, with the 30,322 EUR its sale brought in. It takes the 37.95 BTC L_0 has left at 101.16, then 259.64 BTC from the profit target L_1 has just placed at 102.00, and its EUR are spent. The price jumps again, to 102.00. L_0 is now flat and has a profit of 351.87 EUR from the round trip. S_1 sold 303.22 BTC and got back 297.59 BTC, a loss of 5.63 BTC. L_0 closed by selling its BTC and the result is in EUR. S_1 closed by spending its EUR and the result is in BTC.[7] Each tribe delivers the coin it holds and keeps the score in the other. One agent's stop has become the next price move. Three trades in, the market has produced a rally of 2% with no news, no strategy, and no opinion.

- **Tick 369.** L_1's holding time runs out, and it must sell the 5.63 BTC it still holds. However, there is no bid in the book at all. Its order waits 87 ticks for a buyer to appear. That is illiquidity: not a bad price but no price.

Note that, by construction, several events can unfold within a single tick. For instance, there were two trades at tick 146. The tick tape output by the simulation (`.npy`) only records the last price per tick. As a consequence, the market's true dynamics are only visible in event time, one price per trade. This is captured by the event tape (`.npz`). With larger n, every tick has multiple trades and the book becomes dense enough to prevent large jumps within a single tick. However, each tick can compress a hundred prints or more, and the event tape remains the finest record the market has.

For an $n = 1$-market, the dynamics are a single deterministic ping-pong: one round trip through the two exit levels, two to four trades in total, after which the market freezes forever. One side of the book empties, the lone counterparty is stuck in a position it cannot close, and with no crowd there is no third party to break the deadlock. Interesting dynamics emerge when an agent consumes the exit of another agent of its own tribe, which requires $n \geq 2$.

Emergent dynamics. Everything the large market does is identical to the $n = 2$-market, repeated thousands of times a tick. Prices move when orders are filled and every resting order is somebody's profit target. A stop-loss is a forced trade that moves

[7] For the purpose of this example, accuracy was limited to two decimal places. This incurs a rounding error of 0.3 EUR for S_1. The real calculation holds to 10^{-5}.

the price, possibly triggering a cascade of stops. A time-out into an empty book is a trade that cannot happen. With 15,000 agents, these events overlap so densely that the book is never empty and the gap between the best bid and the best ask shrinks to thousandths of a percent. Nobody is paid to quote and nobody intends to provide liquidity. Such engineered liquidity is an emergent property of this CAS.

Fig. A4 depicts snapshots of a trading dashboard for NNNN and n = 7,500. The price evolution and the corresponding LOB dynamics are shown. The book is never symmetric. Both tribes keep entering as the price moves: those betting with the move rest their take-profits one band ahead of it, those betting against it one band behind. The exits ahead are laid afresh at every level the price passes, while the exits behind recede as it moves away. As a result, depth accumulates on the side the price is moving towards while the other side thins out—a bow-wave emerges. The pile-up is what slows the move: every stop that fires is a market order, and it lands in an exit laid a moment earlier. The thin side is then what ends it: once the opposite direction is cheap, a few agents re-entering on their own clocks are enough to carry the price through the first stops on that side, and the cascade reverses. The book flips with the price, the old bow-wave becomes the brake on the new move, and for a moment the two tribes' stale exits stand far apart with nothing between them. Trends, reversals, and the spread are all produced by the same four rules, with nothing in place to make a market.

The price has no built-in direction. This is seen in Fig. A5, which shows seven different seed runs to 400,000 ticks. Of these, three end above the starting price, three below, and one about where it began. However, the tribes' wealth shares stay at 0.50 ± 0.001 throughout. This balance is not luck: the model is invariant under swapping the two coins together with $x \to 1/x$, so for every upward path there is an equally likely mirror image downward. Symmetry of direction is a consequence of the construction, and checking different seeds merely confirms that the code obeys it.

Within a single run the market produces booms, crashes, and recoveries, alternating between calm stretches and wild ones, with nothing external to trigger any of them. Two runs illustrate the range. Seed 202 falls 91% from its starting price by tick 400,000 and never stops trading. Seed 9, continued to one million ticks, first rallies 220%, then collapses to 92% below its start by tick 650,000, and stays at that depth for the remaining 350,000 ticks. See Fig A6. It does not freeze: orders keep arriving, both sides stay quoted, prices keep forming, and no agent has been excluded. The wealth distribution has barely moved. But it is a different market from the one before the collapse: several hundred longs are stranded in positions whose exits find no buyer, and the sawtooth of the small-n regime is back in the price. The rules did not change, the state did.

Every result of the model is therefore a distribution over seeds, not a number. The same configuration is chaotic in detail but structurally stable: two runs never agree on when a crash comes or which way the price breaks, yet they agree closely on the spread, the tick-scale volatility, the kurtosis, and the scaling exponents.

Stylized facts. These are simplified, generalized empirical observations about exchange rates that hold consistently across different currencies, markets, and time frames [65, 66, 58, 59].

Returns are unpredictable, except for one thing. The correlation between one price change and the next is negative and strong at the tick scale (-0.35 at lag one) and vanishes beyond it. In event time the same negative correlation is spread over the first two prints and dies out within about ten, a tenth of a tick. Real markets show a milder version, called the bid/ask bounce: prices alternate between the buyer's and the seller's side of the spread. In the null model, the effect is larger because every move eats a resting profit target and is immediately pushed back by the next one; half of all trades print at exactly the previous price. This is the market's self-damping. See Fig. A7.

Volatility clusters. Big moves are followed by big moves and calm periods by calm periods, in real markets and here. The autocorrelation of the *size* of returns, ignoring sign, is 0.29 at one tick and decays to zero within about sixty ticks. In event time it decays slowly across four orders of magnitude of prints. Real markets show longer memory still, extending over months. One component real markets have that this model does not is clustering of *activity*, i.e., the bursts and lulls in when trades happen. In the 15,000-agent market every tick carries a trade, so there is no lull to cluster. See Fig. A8.

Fat tails thin out under aggregation. The excess kurtosis[8] of the null model, measured trade by trade, is about 80. Aggregated to one tick, meaning about a hundred trades, it is 1.7–2.3. Over a few thousand trades it falls to zero and the distribution is Gaussian. Real markets show exactly this deflation with aggregation, from intraday to monthly, though their intermediate-scale tails are heavier (kurtosis of 10 and above at the minute-to-hour scale). However, note that the null model's trade-level value is microstructure, not tail risk: about half of all consecutive prints share a price, so the distribution is a spike at zero with a spread of jumps around it, and the excess kurtosis measures that microstructure. Two of the six calm seeds contain a single 3–4% jump, and in windows of a thousand trades or more that one event dominates the statistic until it is averaged away. The crash seed's excess kurtosis stays in the hundreds at every scale. A regime change is not a fat tail, and the statistic cannot tell them apart. See Fig. A9.

The book leans into the move. The lean is the bow-wave mentioned earlier, the exits of the agents who entered during the move, and it predicts the turn: a book heavier on the bid side within 1% of the mid is followed by a rise, a book heavier on the ask side by a fall (57% against 38% at a horizon of a hundred ticks). Real order books show the same sign: the imbalance between the bid and ask queues is the best-known short-horizon predictor of the next price move [67], and changes in that imbalance move the price as they happen [68]. In real markets the signal comes from the two queues at the best bid and the best ask. In the null model those two queues carry no signal at all; it sits further

[8] Kurtosis is a statistical measure that describes the "tailedness" of a distribution. In other words, how likely extreme outliers relative to its average value are. Excess kurtosis is the kurtosis of a distribution minus 3, the kurtosis of a normal distribution. A positive excess kurtosis indicates a distribution with heavier tails than a normal distribution, while a negative value indicates lighter tails. In markets, a positive value means that extreme moves are far more common than a normal distribution would allow.

into the book, in the resting exits behind the best quotes. The real-market predictor is about who is about to trade; the null's is about where the crowd has agreed to stop. See Fig. A10.

Scaling laws in event time. Two particular scaling laws are relevant, see App. C. The number of directional changes, how often the price reverses direction by more than a threshold δ, and the average overshoot, how far each move runs on after the reversal has been confirmed. In real foreign exchange (FX) markets the number of reversals scales like a random walk ($N \propto \delta^{-2}$) and the average overshoot equals the threshold. The null differs on both counts, and the difference is informative. Directional changes fall off much faster ($\delta^{-3.5}$ over 0.4–5%, steepest at the exit-band scale and near δ^{-3} above it). Overshoots carry about half the threshold above the exit band and a fifth to a quarter below it. Moves in the null model systematically fall short. This is the same self-damping as the negative one-lag correlation, seen at larger scales. It is the model's one systematic departure from real markets, and it names what the crowd lacks: nothing in it carries a move further than one band. Seed 202, the collapse, is the instructive exception: its directional changes count falls off like $\delta^{-2.4}$, the FX-like value, made by the sawtooth rather than by anything resembling a market. Any enhancement to the model should also be evaluated in terms of how these scaling laws change, either in quality or realism. See Fig. A11.

The point of this comparison to stylized facts is not that the null model is a real market. It is that these features, when observed in a real market, are not by themselves evidence of information, strategy, or skill. A market of agents with none of those produces them from the order book dynamics alone.

Comparison to the literature. Two benchmarks make the null model's position clear. One is found in the (algorithmic) agent-based tradition, the other from quantitative finance (i.e., analytical).

A classic agent-based market is that of Lux and Marchesi [69]. Its agents come in two kinds, fundamentalists who trade towards a known fair value and chartists who follow trends, and they switch between the two camps according to which is currently more profitable and how many others have already switched. In other words, it is a herding mechanism. The price adjusts to excess demand through a market-maker rule and there is no order book. The model's seminal result is that fat tails and volatility clustering emerge from these rules, with calm periods when fundamentalists dominate and turbulent ones when chartists take over. The null reproduces both stylized facts with none of the ingredients: no fair value, no strategies, no switching, no herding. It does not follow that those ingredients are irrelevant in real markets, however, it follows that the stylized facts cannot be cited as evidence for them, because a market without them already shows the signature. This is what a null model is for. Where Lux and Marchesi attribute the facts to the psychology of switching, the null model locates a sufficient cause one level down, in the mechanics of exits meeting an order book.

A classic analytical benchmark is the Glosten and Milgrom model of a market with informed and uninformed (or noise) traders [70]. A single market maker quotes a bid

and an ask to a stream of traders, some of whom know the asset's true value and some of whom trade randomly. The market maker cannot tell them apart, so every trade is treated as information: a buy raises the probability that the value is high, a sell lowers it, and the quotes are updated after each trade by Bayes' rule. Thus, the prior belief is combined with the likelihood of the observed trade, given the possible values, resulting in a revised belief. The spread is the market maker's protection against informed traders. It is set so that what is lost to those who know is recovered from those who do not. Three things follow in that model. The spread exists because of information asymmetry and would vanish without it. Then, the price is the market maker's best estimate and therefore has no predictable drift. Finally, the spread widens when informed trading is more likely. The null model inverts the first: it has a tight spread with no information, no market maker, and no value to be informed about, because the spread is the geometric residue of thousands of profit targets resting at slightly different levels. It fails the second: the null's price is not an estimate of anything, and it is systematically pushed back after every move (the self-damping that makes it exploitable). And it cannot address the third at all, since it has no informed flow to respond to.

The two comparisons unveil the same finding from different perspectives. The null model reproduces fat tails, volatility clustering, a tight spread and the bow-wave with no cause beyond the mechanics of the order book itself. The agents have zero-intelligence and are identical except in wealth, timing, and luck. Observing these properties in a real market is therefore not, by itself, evidence of information, strategy, or skill. What the null model cannot reproduce is any response of the spread to the flow of trades it faces. No agent reads the flow, so there is no adverse selection and, thus, no defense against it. Nothing in the book would protect a resting order from toxic flow that systematically takes money from whoever rests there, and nothing would ever widen a spread. Both responses are exactly what an informed agent would bring. In the language of market microstructure the sustainable spread of a market-making engine is the one that pays for adverse selection, e.g., [70]. In the Liquidity Engine this spread is not calculated but found: the state in which what the market-making tribe loses to informed flow is covered by what it earns from the rest.

The closest precedent to the model's functionality is the zero-intelligence tradition. For instance, Gode and Sunder's budget-constrained random traders, which reproduced the efficiency of double auctions [71]. In other words, the ability of a such markets to perfectly match the right buyers with the right sellers. Another example is Farmer, Patelli, and Zovko's random order flow, which predicted the spread and price impact of the London Stock Exchange with no strategy at all [72]. Those results established that the mechanics of a market can explain more than its participants' intelligence does. The null model takes that separation one step further. In an ABM, an emergent feature can have three sources that are routinely conflated: a *convention* of the modeler (the numeraire, how percentages are computed, what a close delivers, in which order agents act), the *mechanics* of the market (the order book, the position life cycle, the conservation of money), and the *intelligence* of the agents (information, strategy, learning, cooperation). Farmer et al. separated the third from the second. Nobody, to our knowledge, has separated the first from the second, and Sec. 3.2 showed how what looks

like market behavior turns out to be the modeler's bookkeeping, and Sec. 3.3 what the bookkeeping has to be for the game to be fair. The null model is a control that holds all three apart: conventions made symmetric so they contribute nothing, mechanics kept, intelligence absent. Whatever it produces is mechanics; whatever it cannot produce is the remaining budget for intelligence.

In summary, the symmetric market is artificial by design, as any control must be. The claim is not that real markets are symmetric but that a real market's asymmetries—tick size, quote currency, fee schedule—are treatments whose effects can be measured against a baseline that has been shown to contribute none of its own.

The order book. Nobody in this model is paid to quote, and no agent ever places an order for the purpose of providing liquidity. Yet the large market is continuously two-sided: over the seed battery, a bid and an ask both rest within 1% of the mid-price in every snapshot of the calm market, and the median spread is 0.03–0.04% of the price, some thirty times smaller than the exit band that generates it. The reason is arithmetic rather than intention. Every open position leaves a take-profit resting one band from its own entry price, and 15,000 agents entering at thousands of different prices leave thousands of targets at thousands of different levels. The book within 1% of the mid typically holds several hundred distinct price levels; the price axis is paved almost continuously, and the spread is simply the gap between the two nearest tiles. Recall Fig. A4.

This paving is a crowd product, and the crowd's size is what sets its quality. At 75 agents per side, with the same total capital, the median spread is 0.40% and the book is two-sided only 84% of the time; at 250 per side, 0.06% and 93%; at 750, 0.06% and 99%; at 7,500, 0.03–0.04% and 100%. See Fig. A12. The first tenfold increase in head count tightens the spread sevenfold and closes the gaps in coverage, the second only halves it. Once the price axis is paved, more hands holding the same capital add depth to the tiles rather than new tiles—and with no smallest price step, what the spread then measures is how finely the crowd happens to disagree, not how much it is prepared to trade.

With no minimal tick size, nothing bounds the spread from below except the number of agents quoting near the best bid and ask. The null model's calm-era spread, 0.04%, is already below the one-cent step a 100-EUR instrument carries on any real venue, and a third of all snapshots sit below a cent. The collapse of the spread shows what that number is worth: as one tribe strands, the median spread falls tenfold to 0.004%, because the few quotes left on each side sit at the best prices with no depth behind them. The tightest spread the null model ever quotes is its anomalous state. A spread, in other words, is only informative against a price grid: the first real-market convention to add is a tick size, symmetric as a log-price grid, which turns "how many agents" into "how much depth at each price" and gives the spread a floor that means something. See Fig. A13.

Depth and coverage are set by different things. Coverage—how many levels are populated and how often both sides are present—is a function of n. Depth—how much can be traded before the price moves—is a function of K, and the order size divisor q. Note

that rescaling the total capital by a factor of eight reproduces the same price path with every resting volume exactly eight times larger. A bigger crowd does not make a heavier book, because the crowd's capital is fixed, and more agents merely slice the same wall finer. Thus, more agents buy quality, and more capital buys capacity. For the Liquidity Engine this is the operative distinction: the quality of the book is demonstrated at zero cost, by the mere presence of a crowd, while its capacity is a funding decision.

Depth is not evenly distributed around the price either. It leans toward the side the price is moving towards, the bow-wave of Fig. A4, seen here in numbers: after a rise over the preceding fifty ticks, the resting value within 1% of the mid on the ask side exceeds that on the bid side by 10–15%, after a fall the reverse, and the lean decays over a couple of thousand ticks (Fig. A10). It is the same mechanism as the tick-scale mean reversion of the stylized facts, viewed from the book's side: everyone who bought on the way up parked a take-profit above their entry, so an advance lays a wall of its own profit-taking just ahead of itself and climbs by consuming it. Note that the resting volume counted in BTC rises as the price falls only because the same EUR buy more BTC. So, counted in value, the book changes far less through a large swing, and the value measure is the one that means anything to a trader.

Market longevity. How long does the market hum last? One run, seed 17, was continued to 2.2 million ticks to find out. See Fig. A14. The answer is about a million ticks, and then pathologies appear. For the first 400,000 ticks the market is healthy. It then rallies to three and a half times its starting price, falls to a tenth of it, and spends the next 400,000 ticks at depth. Through all of this, wealth stays almost as evenly shared as at the start: the richest 1% of agents own 2–3% of everything, and the effective number of agents—the number that would share the wealth if all shares were equal—stays above 10,000 of 15,000 for the first 870,000 ticks. Nobody has gone broke, and nobody will, and at the end of the run every one of the 15,000 agents is still trading.

Then the wealth distribution breaks, in two stages. From tick 870,000, the effective number of agents starts to slide, and it is 5,400 at tick 1.0 million. Then, between 1.1 and 1.2 million the market rallies 25-fold and the number falls off a cliff, to about 100. From then on the richest 1% own 29% of everything, and a few dozen agents effectively own the market. The mechanism is not that anyone trades better than anyone else. It is that a rally multiplies the wealth of whoever happens to be holding the asset at the time, and after a long period of low prices a fraction of the buyers were holding a great deal of it. A run of luck, compounded by a large move, is enough to create whales.

Once the whales exist, the market acts as a small one. A hundred effective participants is the size at which, described in Sec. 3.3, the small-market pathologies appear: for the remaining million ticks the price swings between half and twice its new level, hundreds of agents are periodically stuck in positions they cannot close, and the crowd enters and exits in waves. The number of open positions per side alternates between a few hundred and over a thousand. The market does not die. It degrades into a large-looking market with the dynamics of a small one.

Two things to note. This is one seed. The two others (seeds 9 and 202, to about

850,000 ticks each) show the same slow concentration through their collapses with the effective number down to about 11,500, and the richest 1% up to 3%. However, neither was continued long enough to see whether whales form. And the timescale is not fixed: concentration is driven by large moves, so a run that never rallies or crashes may hum far longer. What the run establishes is the fate, not the date. A market with selection and no renewal ends up owned by the lucky.

Exploitability. Consider a rule that trades against every directional change: each time a 1% reversal is confirmed, it takes a position against the new direction, and holds it until the next 1% reversal is confirmed, at which point it closes and takes the opposite position. Its profit on each trade is the threshold δ minus the overshoot ω, so the expected profit is $\delta - \langle\omega\rangle$. In a real market, where $\langle\omega\rangle = \delta$ [59], that is zero.

In the null model, $\langle\omega\rangle$ is a third of δ at the 1% threshold and about half of it above. The distribution is skewed: at 1% the median overshoot is a tenth of δ, one in seventeen exceeds δ, one in two hundred exceeds 2δ. Moreover, the tail is thinner: the rare large losses no longer offset the many small gains. The reason is who is in the market. Every resting order is a take-profit, a limit order leaning against the move that reaches it, and the book holds one every thousandth of a percent. Nobody in the crowd ever sends an order that continues a move. A real market has the same dense book and, on top of it, participants who carry moves on—momentum, information, herding—and the two balance to $\langle\omega\rangle = \delta$.

The null model has only the brake, so the expected profit of the rule is positive: 87% of its trades win, about 70 basis points each, before costs. However, how much an agent can profit depends entirely on how fast it is. Filled at the trade after the signal, net of the spread, the rule keeps about 20 basis points; two trades later, 9; five trades later, nothing. One tick later, when the reversal it is betting on has already happened, it loses 13. The edge is real, it is one to five trades wide, and across forty thousand reversals in 400k ticks it would drain the crowd.

Nothing in the null can prevent exploitation. No agent reads the flow, so the crowd cannot respond to anyone who does, neither to this exploiter nor to an actor with real information. As a result, the market's spread would not widen and its depth would not retreat, and nobody is faster than the exploiter, because nobody is trying to be. Wealth drains without resistance.

Summary The zero-intelligence null model, with all its functionality dictated by symmetry constraints, produces a liquid market[9], and most of the stylized facts of real markets. However, it is also exploitable and pathologies appear when the market is left to run for long enough and whales emerge. Thus, the model is complete as a control and incomplete as a venue. The next step is to add realism and intelligence. A foundational idea is to incorporate intrinsic time. Before addressing this topic in Sec. 3.6, the null model is made accessible as a tradable exchange.

[9]Recall that the spread and the LOB depth are controlled by only two parameters. The spread size diminishes with n, and the depth is a function of the available capital K.

3.5. The Null Model as a Trading Terminal

A working proof-of-concept is provided by incorporating the null model, unchanged, into a browser-based trading interface for a fictitious currency pair, RCC/EUR, opening at $x_0 = 100$. In essence, this is a simulated venue. The GUI can be accessed at `https://jbglattfelder.github.io/alphaEngine/venue/`, with username `liquidity` and password `engine`. The source code and README are found in the `venue/` directory of the public GitHub repository. To increase the number of quoting levels, increase n (limited to 100,000 agents per side). To deepen the book, increase the capital K or decrease the order divisor q, since the resting volume is expected to scale as K/q ($q = 1$ is an agent's entire wealth per order). See Figs. A15 and A16 for screenshots.

The engine runs entirely on the client: the Python code is compiled to WebAssembly, so every visitor operates a private market and no server is involved. Before trading starts, the user chooses the configuration (NNNN or PFCF), the population size n (up to 10^5 agents per side), the capital K, the exit bands, the clock rate c, the order fraction q, the random seed, and their own endowment of EUR and RCC (on top of K, at most $0.75\,K$). The human trader enters the market through the same order book as the agents: market and limit orders are matched with price-time priority at the makers' prices, a resting limit order supplies liquidity that the agents consume, and, because there is no market maker, the bid-ask spread remains emergent. Profit and loss are computed with the balance-sheet formalism of [64], positions being marked at the side of the book against which they would be closed. A short sequence of trades suffices to experience the null model's microstructure first-hand: liquidity vanishes and reappears tick-by-tick, a single order can sweep one side of a thin book, and a limit order far from the last price is filled whenever it becomes the only quote on its side. Two checks show that the terminal leaves the dynamics untouched: in the absence of human orders, the price path coincides bit for bit with that of the engine run natively, and a regression test reproduces the tick tape of two reference markets ($n = 2$, both configurations, 2×10^4 ticks) exactly in the browser before each release.

3.6. Enhancing Fidelity: Intrinsic Time

One key drawback of the null model's agents is that they all share the same clock. In other words, the activity-adjusted behavior of intrinsic time is missing. This is the first layer of realism to add.

As every agent in the null model acts on the same clock, its pressure to trade rises by the same amount each tick, whether the market moved or not, so a quiet market and a turbulent one are traded at the same pace. Real participants do not act that way: they react to events. The first extension therefore gives each agent its own intrinsic clock, which advances only when the price has moved or reversed by that agent's own threshold. These are the directional-change or overshoot events of Sec. 2.5. Nothing else changes.

Intrinsic time clocks thus add an additional degree of freedom to the dials, labelled I. In detail, this affects the entry clock (capital, 1st dial), the time-out (closing, 3rd dial),

and the order size (4th dial). Recall that in the null model an agent's capital sets how often it enters, because pressure accrues per tick and a wealthier agent fires sooner. The capital dial in I keeps the capital draw but lets the pressure accrue due to intrinsic time events instead. Thus, the entry clock runs on the agent's threshold rather than on ticks. Furthermore, note that the size dial has no clock of its own. In I mode, it reads the same threshold the two (entry and exit) clocks tick on, so that an agent whose clocks run on small moves trades small and one whose clocks run on large moves trades large. For this reason, NNNI cannot exist. Also observe that calibrating the exit bands (2nd dial) with intrinsic time would actually introduce agent behavior, as the market state would inform trading decisions. Therefore, we don't include _I__ in this analysis. However, to add even more realism without introducing intelligence, the exit band thresholds can be drawn from a heavy-tailed distribution (P for Pareto) instead of a normal one, so that the crowd holds scales from a third of a percent to fifteen. As a result, two maximal intrinsic time arms are possible: one impacting the three dials called INII, and one with all changes, called IPII. Note that a Pareto draw on the size dial (I_IP) is omitted, as it would introduce a second, independent source of heterogeneity. In I_II the size diversity is inherited from the exit thresholds, so any intrinsic time change relative to the null can be attributed to one ingredient.

A scan switched the intrinsic time clocks on, one dial at a time, at $n = 500$ per side, a size at which the null model is mostly pathological (recall Fig. A2). Putting the entry clock in the agent's intrinsic time (i.e., INNN, INNI, and ININ) removes the sawtooth on every seed and keeps the price within e^2 of its start on all of them (the null model's range spanned e^6, and it locked against its wall in five seeds of six). The time-out is no help, alone or together with the order size (i.e., NNIN and NNII): the price still runs away on every seed, and can reach e^6. The reason is that the time-out only decides when a position is abandoned, not when the next one is opened, and it is the entries that pile up on one side. Putting all of the agent's possible dials in intrinsic time (i.e., INII) gives the calmest arm: the price ends within $e^{1.04}$ of its start on all six seeds. How the thresholds are distributed—normally (N), heavy-tailed (P), or identical for everyone (F)—changes little, which also retires the prediction that identical thresholds would lock the crowd into step. IPII is the attempt to incorporate the most realism. It keeps the three intrinsic dials, replaces the normal exit bands with Pareto ones. Note that it ties the two scales together: the exit band an agent draws is also its event threshold, so each agent has one scale for take-profit, stop-loss, and its clock. As a result, the crowd spans scales from a third of a percent to fifteen. This puts motion back. The trade count rises by a third over INII's, the price moves again but stays within a factor of e of its start on five of the six seeds (seed 202 ends at $e^{2.1}$). As the next paragraphs show, INII introduces new pathologies, while IPII is the arm with the tightest spreads, similar stylized facts to NNNN, partial scaling laws, and market longevity. See Fig. A17 for the corresponding plots.

Figures A18–A21 summarize three seeds of IPII against the INII control for an extensive run. The price (Fig. A18) stays diffusive and well-behaved, drifting 2–11% per 400k ticks, with no runaways, and no visible difference between seeds. The stylized facts (Fig. A19) show the differences of the two arms. IPII matches the null model better:

the bounce sits near -0.35, fat tails wash out under aggregation, and the book leans ahead of the price out to 10^4 ticks. Only the volatility memory differs, at 20–30 ticks against the null model's 60. INII diverges: tick kurtosis of 10, volatility memory of 200–300 ticks, and entry avalanches whenever the crowd's shared 1% scale is touched. The scaling laws (Fig. A20) tell the same story in event time: IPII follows $\tilde{N} \propto \delta^{-2.7\ldots-3.0}$ between $\sim$0.9% and $\sim$3.5%, and its overshoot holds $\langle\omega\rangle = \delta/2$ from $\sim$0.7% to beyond 4%, where the null model obeyed its steeper law only above its single band scale. INII's event-time scaling is pathological, for both directional changes and overshoots: they obey a law only in short segments.

The book is where IPII clearly improves on the null model (Fig. A21). It is two-sided in every snapshot, stationary from the first window, identical across seeds to the fourth decimal, and paved so finely that its healthy median spread of 0.0035% sits at the level the null model only reaches in collapse. Looking at INII's spread, it never stops falling over the entire run—it is a market that never reaches a steady state. The comparison isolates the ingredient that matters, because the two arms bracket the null model. INII and the null model share the same single exit band and differ only in their clocks, and switching the clocks alone makes the market worse—not because event-driven clocks are harmful in themselves, but because they couple every agent's timing to the price. With the whole crowd listening at the same 1% scale, one move advances every clock at once, and the result is avalanches at that scale and stillness everywhere else. IPII and INII share the same clocks and differ only in the distribution of their exit bands, and this repairs everything: with exits spread from 0.3% to 15%, no single move can trigger more than a sliver of the crowd, each band has its own group of active agents, and the scaling laws now reach below the single band scale that capped the null model's—the count law holds over 0.9–3.5% and the overshoot law from 0.7% to beyond 4%. Event-driven clocks are thus not an improvement by themselves. They amplify whatever variety the crowd has: one shared scale becomes a pathology, a spectrum of scales becomes structure. The null model shares INII's single exit scale but escapes its pathology, because its agents act on a schedule that ignores the price: the uniform tick clock keeps them out of step by accident. IPII keeps them out of step by construction, as different agents wait for moves of different sizes. Neither ingredient is pathological alone, only price-driven clocks at one shared exit band are. One important observation is that spreads can become zero (but never cross): in the order-book snapshots taken every 50 ticks, the book occasionally locks—in 0.08% of cases for NNNN, 0.24% for INII, and 0.27% for IPII.

In summary, IPII engineers a healthy market (compared to INII and NNNN) with believable statistics. On the event-time scaling laws, where real FX and crypto markets are similar to a random walk ($\hat{N} \propto \delta^{-2}$, $\langle\omega\rangle = \delta$, see [59, 73]), the null model produces a directional change count that falls off too steeply ($\delta^{-3.3}$) and its overshoots run on too short ($\langle\omega\rangle \leq \delta/2$), while INII overall has shorter segments producing scaling. In contrast, IPII shifts the directional-change law towards real markets ($\delta^{-2.7\ldots-3.0}$ over thresholds of 0.9–3.5%), however, with the overshoot still at $\delta/2$. The latter is the result of a feature the crowd still lacks: agents who carry moves on. Notably, all properties are emergent from a system of agents with no information, no strategy, and no idea what anything is worth. The code for the intrinsic time models is located in a private repository:

`https://github.com/jbglattfelder/liquidityEngine` (without credentials the page will return a 404 error).

In closing, it should be noted that while the analysis of NNNN is (mostly) exhausted, IPII still has potential avenues for future research. For one, the minimum of the Pareto distribution could vary, as now many small agents cluster at the same scale. Also, as mentioned, connecting the exit bands to intrinsic time clocks (IIII) should introduce novel dynamics through agent intelligence.

4. Conclusion and Outlook

We present a working zero-intelligence agent-based model with longs and shorts trading through a limit order book based on simple rules of interaction, guided by insights from complexity science. The emerging market exhibits bubbles, crashes, trends, volatility clustering, fat tails, and some scaling laws similar to real crypto and FX data. There is no market maker, so the bid-ask spread is emergent: in the calm market at large size, a bid and an ask rest within 1% of the price in every snapshot, with a median spread of 0.03–0.04%. The parameter space is constrained, as there are only four dials: when to open, when to close (take profit, stop loss, timer), wealth distribution, and trade size. Most of the development effort went into making the null model symmetrical. This was achieved by hunting every hidden mechanical bias (random ordering, symmetrical percentages, sizing symmetric at every price, own-currency settlement, no self-trading, etc.). The engine is deployed as a trading terminal that runs in any browser.

In addition, we enhanced the null model by implementing intrinsic time: every agent acts on its own clock, paced by the market's events at that agent's scale. A control arm shows that the clocks alone don't improve the market (agents sharing one exit scale act in avalanches). However, the same clocks with a spectrum of exit scales make the emergent market better on every measure: the null model's stylized facts survive, and the scaling laws move towards the exponents of real currencies. Moreover, the book is two-sided in every snapshot, stationary from the start, identical across seeds, and quotes a median spread a tenth of the null models's. The intrinsic-time markets also live longer: every run survives past where null markets of the same size turn pathological and saw-tooth. The overall aim is a framework versatile enough to engineer different markets that are adaptive and resilient, providing liquidity in any scenario.

The aim of this work is not only to provide a proof-of-concept for a zero-intelligence market, but crucially offer a potential research agenda informed by complexity science. The null model can easily be enhanced by adding information and strategy to the agents, introducing intelligence into the system. Only now is it possible to measure the effects of each addition against the zero-intelligence baseline. For instance, markets can be engineered that implement a homeostatic mechanism, intrinsic price-stabilizing feedback loops, or an external market-tracking arbitrage functionality.

A. Figures

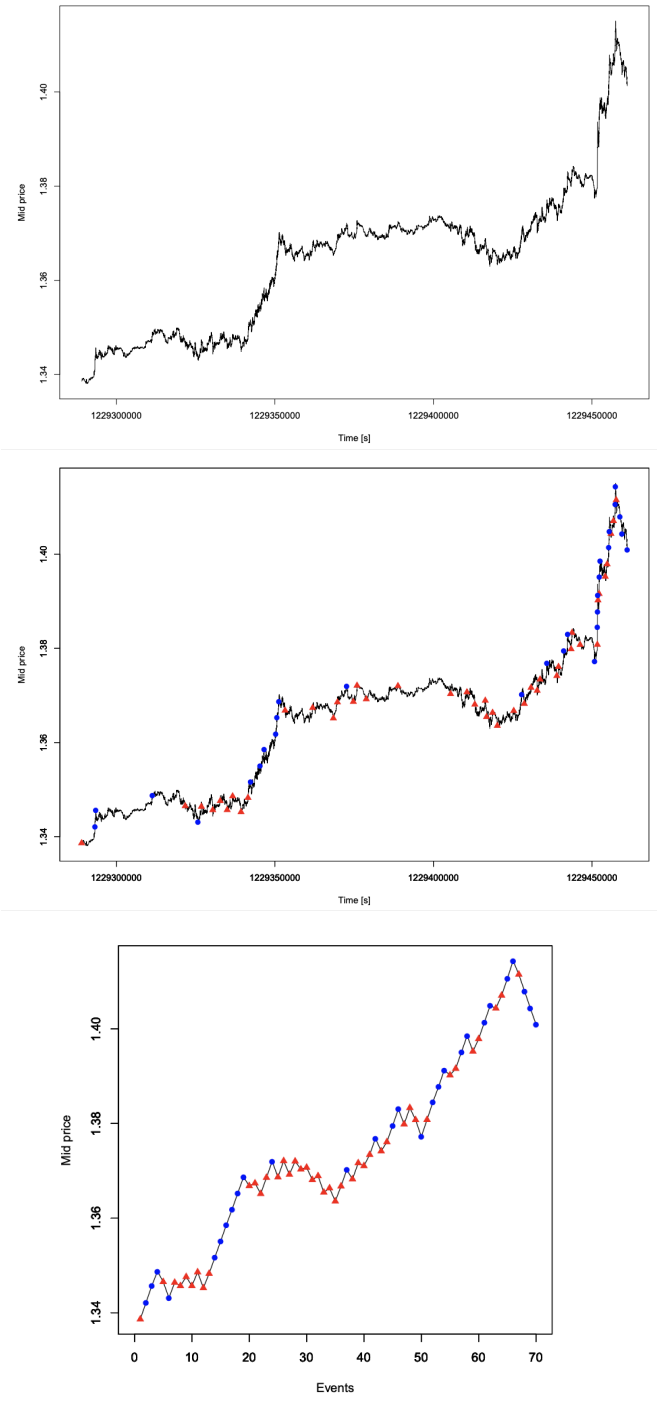


Fig. A1: The intrinsic time operator. Transforming a price curve into its directional changes (red triangles) and overshoots (blue dots) for a defined threshold. Reproduced from [74].

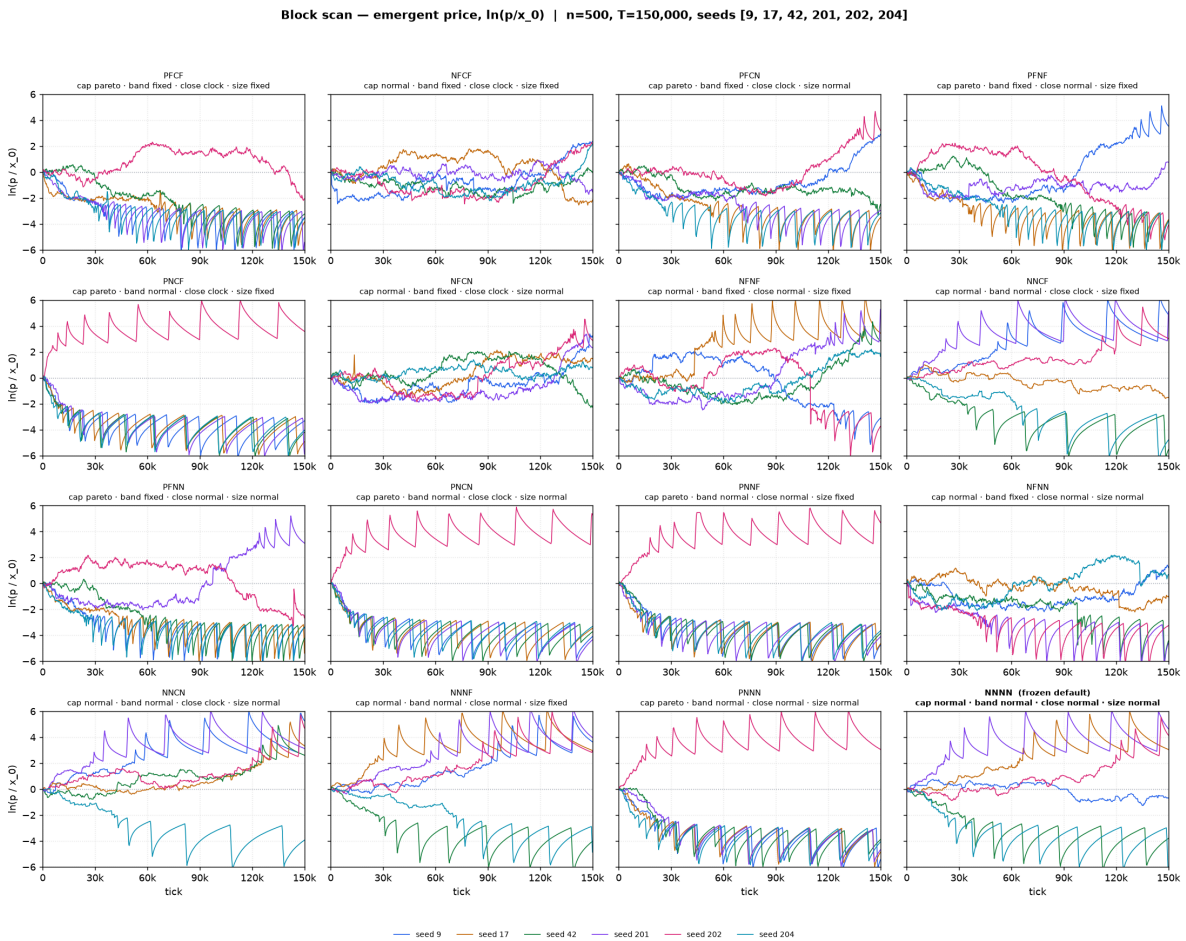


Fig. A2: Dial scan. Price evolution for the 16 possible dial configurations for six seeds.

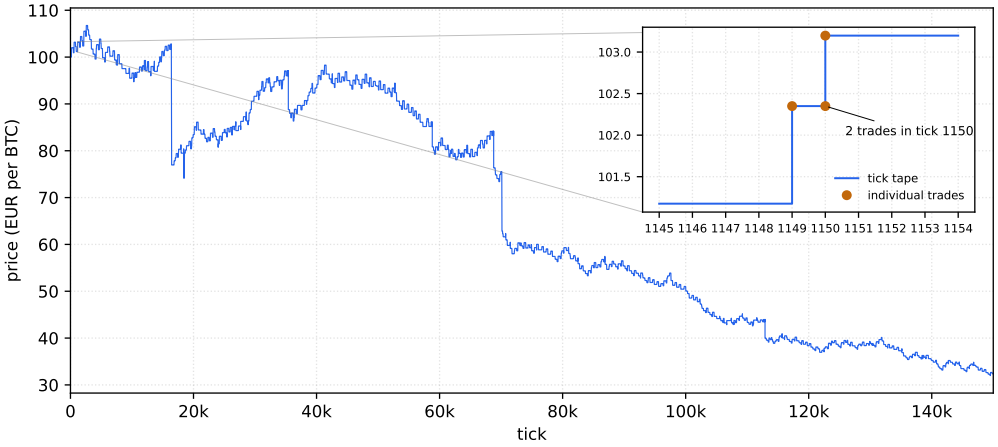


Fig. A3: NNNN with $n = 2$ and $S = 9$. Price evolution of the smallest market. Most ticks are inert, while some contain more than one trade event (see inset).

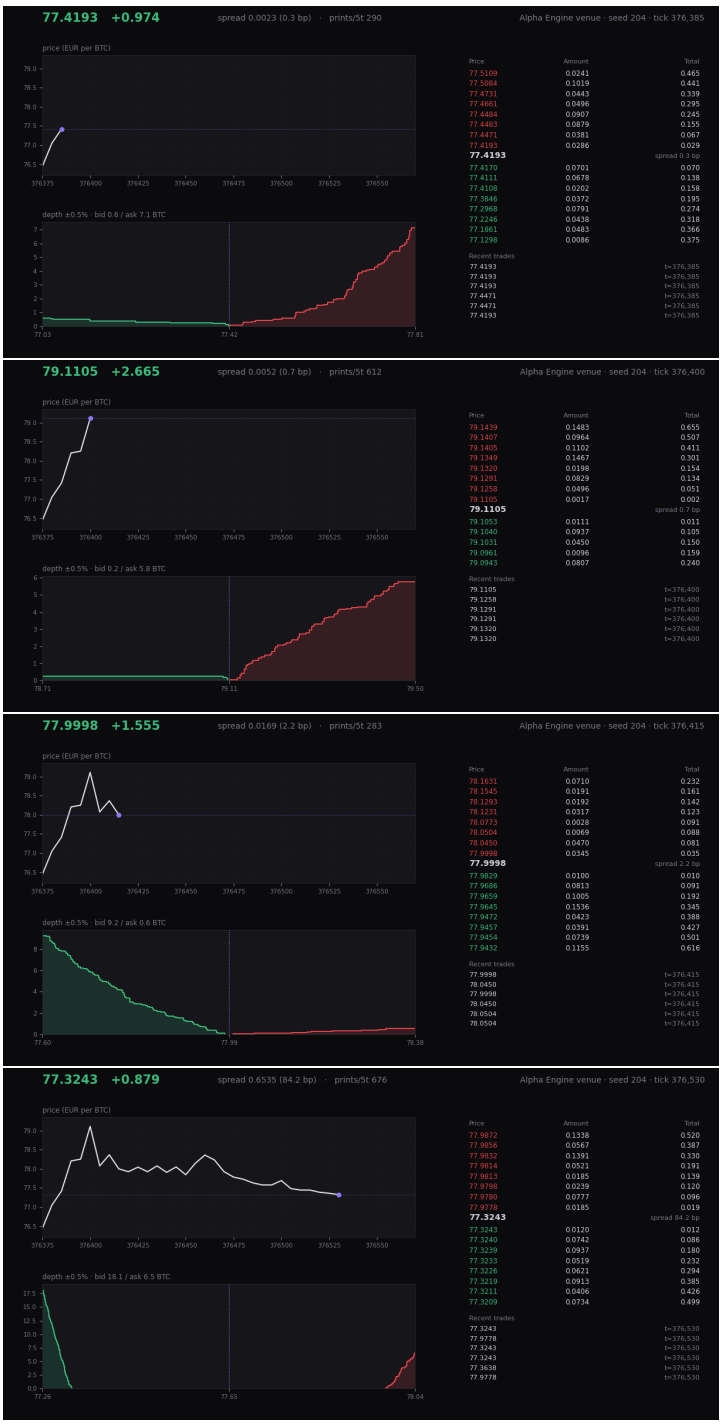


Fig. A4: Trading dashboard snapshots. Price evolution, bid/ask levels, LOB (±0.5%), and trade events for a single run (NNNN, $n = 7{,}500, S = 204$) are shown. Convention: book is end-of-tick snapshot, prices are the last print per tick, and trades are stamped with their tick. 145 ticks are shown, starting from tick 376,385.

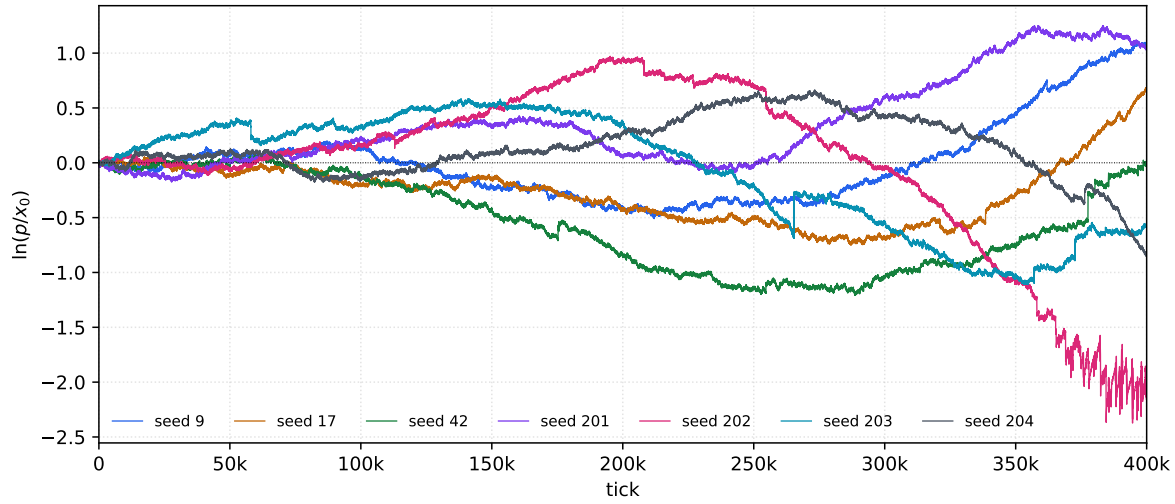


Fig. A5: NNNN with $n = 7,500$. Price evolutions for identical configurations with different seeds.

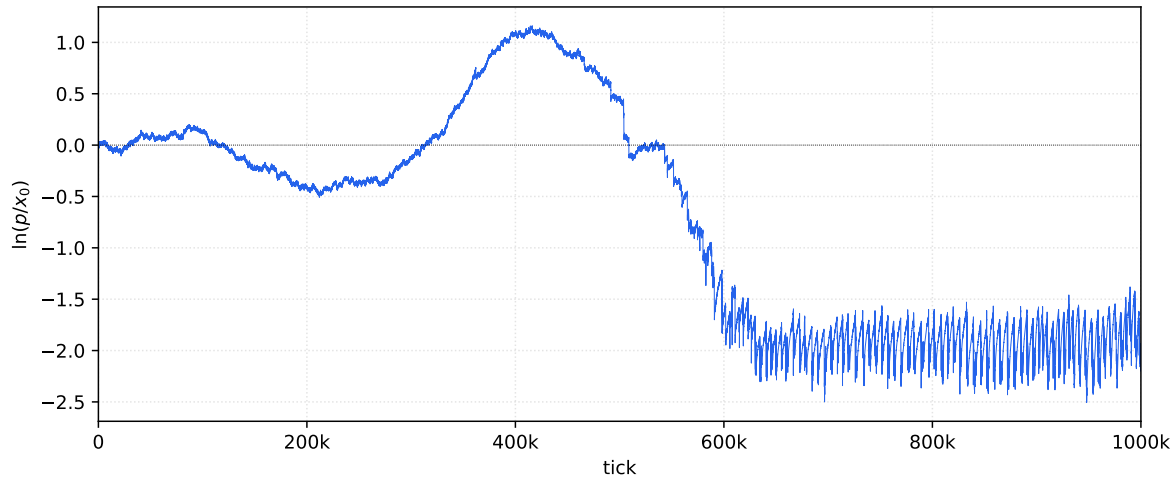


Fig. A6: One million ticks. NNNN with $n = 7,500$, and $S = 9$.

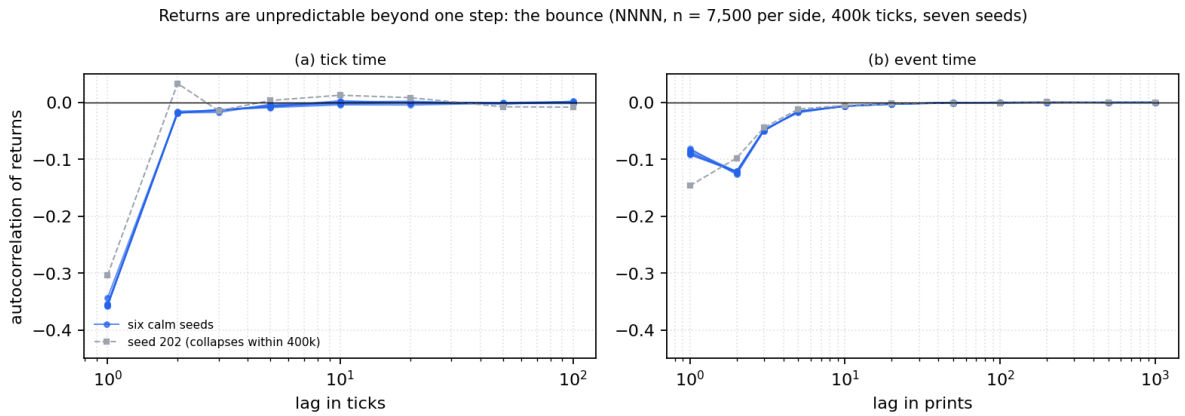


Fig. A7: Autocorrelation of returns. Each point asks: if the price just moved up, is the next move more likely to be up (positive) or down (negative)? (a) From one tick to the next the answer is strongly "down" (−0.35), then zero: beyond one step the past tells you nothing. (b) Trade by trade the same bounce is spread over the first two trades and is gone by the tenth. Blue: the six calm seeds, which lie on top of each other; grey: the seed that collapses.

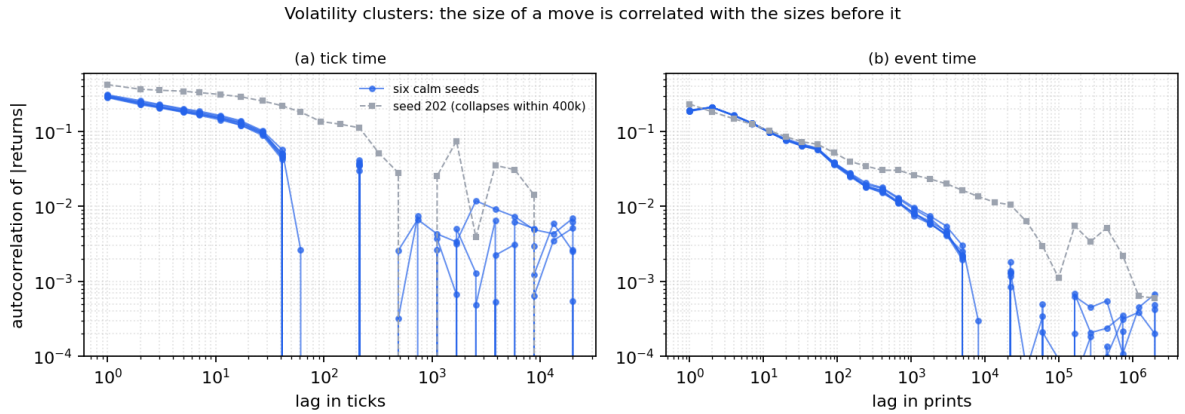


Fig. A8: Volatility clustering. The same question for the *size* of moves, sign ignored: is a big move followed by big moves? Yes, and the memory is long. The curve stays positive out to about sixty ticks in (a) and across four orders of magnitude of trades in (b), decaying slowly rather than switching off. A Gaussian random walk would sit at zero from lag one. The collapsing seed (gray) has longer memory still: the sawtooth is one long cluster.

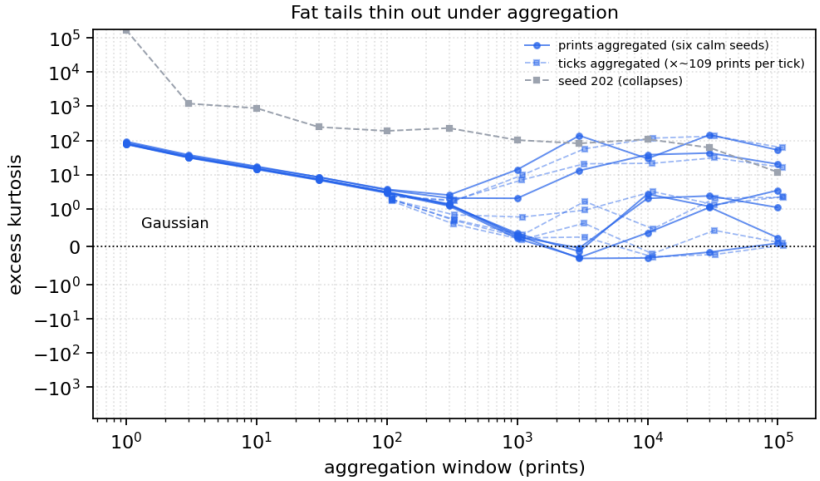


Fig. A9: Aggregation thins out fat tails. Excess kurtosis measures how much more often extreme moves occur than in a Gaussian (which scores 0). Measured per trade (left end) the null model is extremely fat-tailed, around 80. Summed over a hundred trades, or one tick, it is about 2. Aggregating over thousands of trades and it becomes Gaussian. The curves that climb again towards the right are seeds containing 3–4% jumps, dominating any window containing them. Gray curve: the collapsing seed, fat-tailed at every scale.

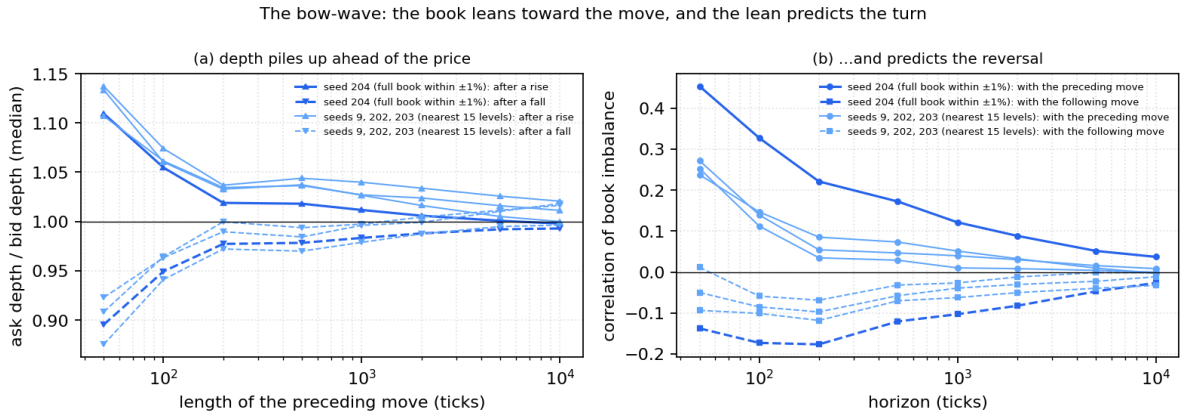


Fig. A10: LOB leans into the move and predicts the turn. (a) After the price has risen over the previous n ticks (up triangles), the resting sell orders within 1% of the price outweigh the resting buy orders by 10–15% for short moves. After a fall (down triangles) the reverse happens. This is the bow-wave: the exits of the agents who entered during the move pile up ahead of it. (b) The imbalance correlates positively with the move that built it (circles) and negatively with the move that follows (squares) at every horizon beyond a hundred ticks. Dark blue: seed 204 (full book). Light blue: three seeds (nearest 15 levels).

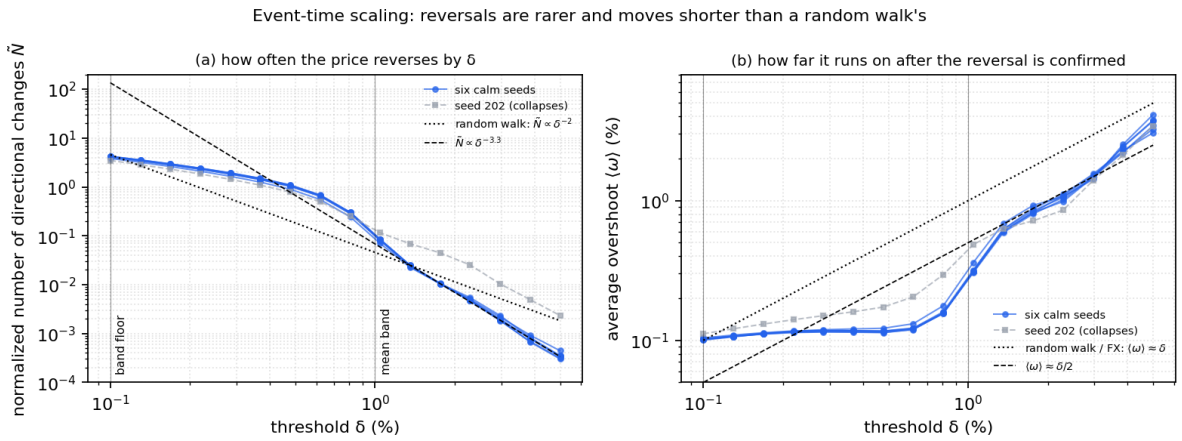


Fig. A11: Partial scaling laws in event time. (a) Number of directional changes (normalized by the time horizon), for thresholds δ from the smallest exit band in the population (0.1%) to 5%. Small reversals are noise: the price turns by 0.1% about four times every tick. Large ones are rare, and become rarer faster than by chance: for a random walk, doubling δ cuts the count by four (dotted). In the model, above the typical exit band of 1%, doubling δ cuts it by ten (dashed). The bend between the two regimes sits where the agents' exit bands are, $1\% \pm 0.3\%$: a move that reaches that size runs into a wall of resting orders. (b) After a directional change of δ, how much further does the price overshoot before turning again? In FX markets, and for a random walk, about another δ (dotted). Here, below the exit band, it is a constant 0.1%, and above about half of δ (dashed). Both panels agree: nothing in this crowd carries a move further than one exit band and scaling is only seen above the mean band of 1%. Blue: the six calm seeds. Gray: the seed that collapses, whose sawtooth imitates a random walk's scaling without being one.

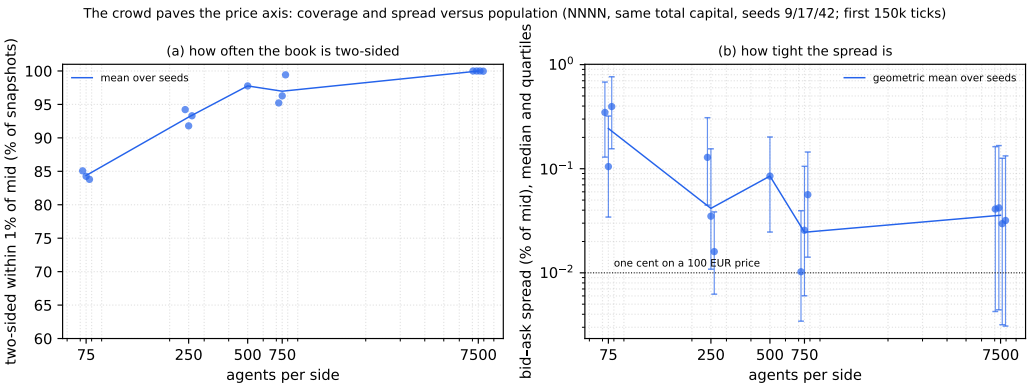


Fig. A12: Coverage and spread versus population. The same total capital, split among 75 to 7,500 agents per side (NNNN; three seeds per point below 7,500, measured over the first 150k ticks, before the small markets run away; four seeds at 7,500). (a) How often a buy and a sell order both rest within 1% of the price, steadily growing with the crowd size. (b) The gap between the best buy and the best sell order (median, with quartiles): it drops tenfold between 75 and a few hundred agents and then stops improving. The dotted line is one cent on a 100-EUR price.

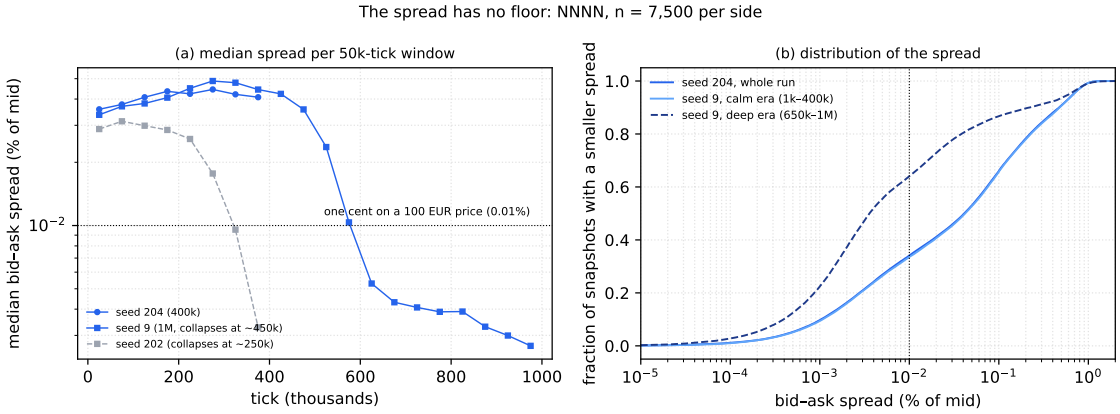


Fig. A13: The spread has no floor. (a) The median gap between the best buy and the best sell order, in windows of 50k ticks, for three seeds. The dotted line is one cent on a 100-EUR price, the smallest spread a real venue would allow. In the calm market the spread is 0.04%, a collapsed market (e.g., seed 202 at ∼250k ticks, seed 9 at ∼450k) it falls tenfold and stays there. (b) The whole distribution: in the calm era (solid) a third of all snapshots are already below a cent; in the collapsed era (dashed) two thirds are. Nothing in the model sets the smallest price step, so the spread is bounded only by how many agents happen to quote near the last price. The collapse produces the tightest spread of all. A tight spread is not by itself liquidity.

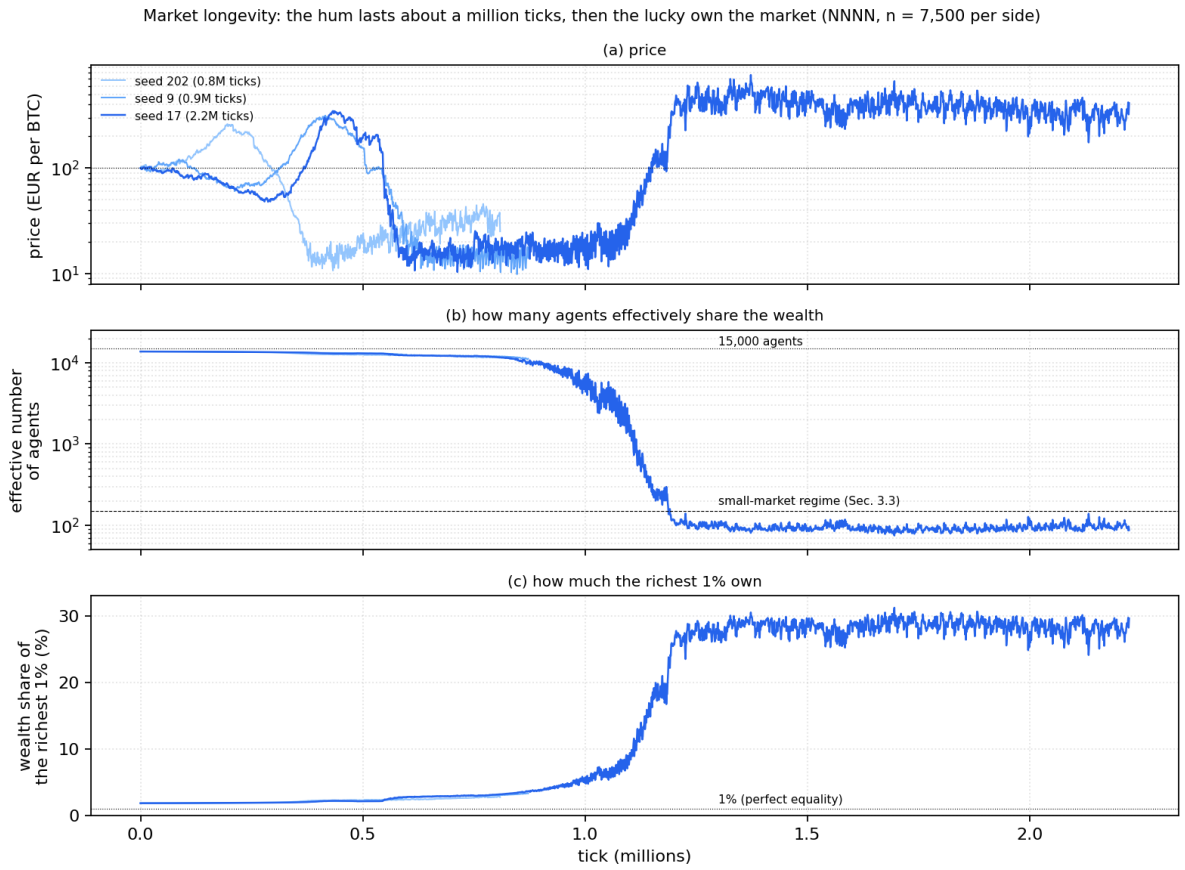


Fig. A14: Market longevity. Three runs of the null model followed far beyond the usual 400k ticks: seed 17 to 2.2 million, seeds 9 and 202 to about 850k. (a) The price. All three collapse to a tenth of the starting price between 400k and 700k ticks and sit there; seed 17 then rallies 25-fold around tick 1.1 million and swings between half and twice that level for another million ticks. (b) How many agents effectively share the wealth: 15,000 would mean perfect equality. The number drifts down slowly through the collapse, is still above 10,000 after 870k ticks, and then falls off a cliff during the rally, to about 100—the size at which small-market pathologies set in (dashed line). (c) The share of all wealth held by the richest 1% of agents: 2–3% for most of a million ticks, 29% after the rally. Nobody has left; all 15,000 agents are still trading at the end. A market with selection and no renewal ends up owned by the lucky.

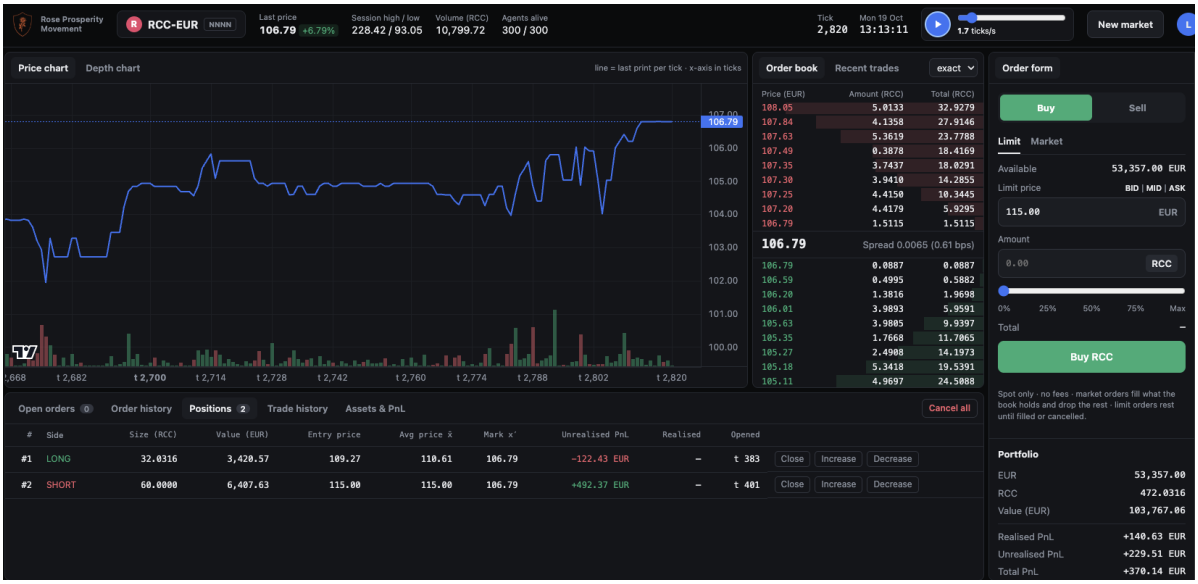


Fig. A15: Trading GUI. Screenshot of a fully functioning trading venue, implementing NNNN and PFCF. Limit and market orders are implemented and positions are tracked. Note that the LOB is a separate tab next to the price chart.

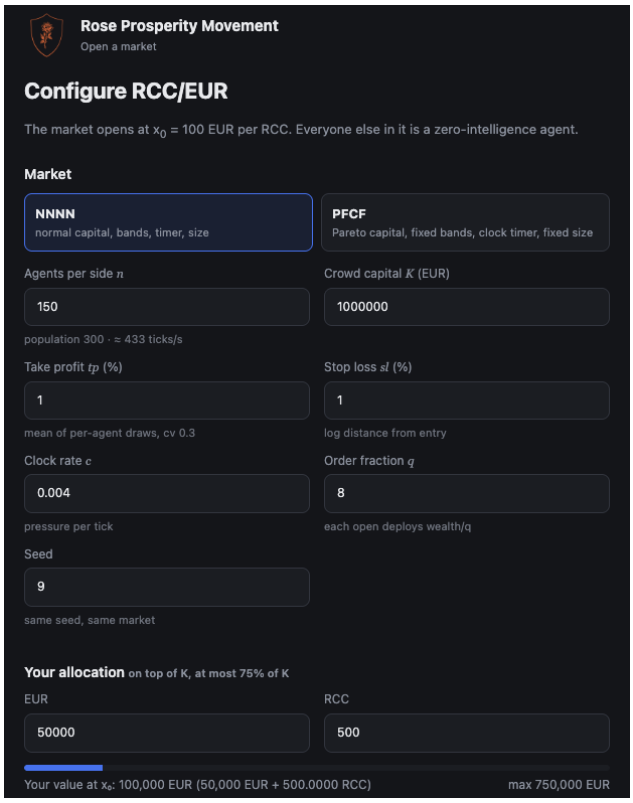


Fig. A16: Trading GUI configuration. Exposed parameters.

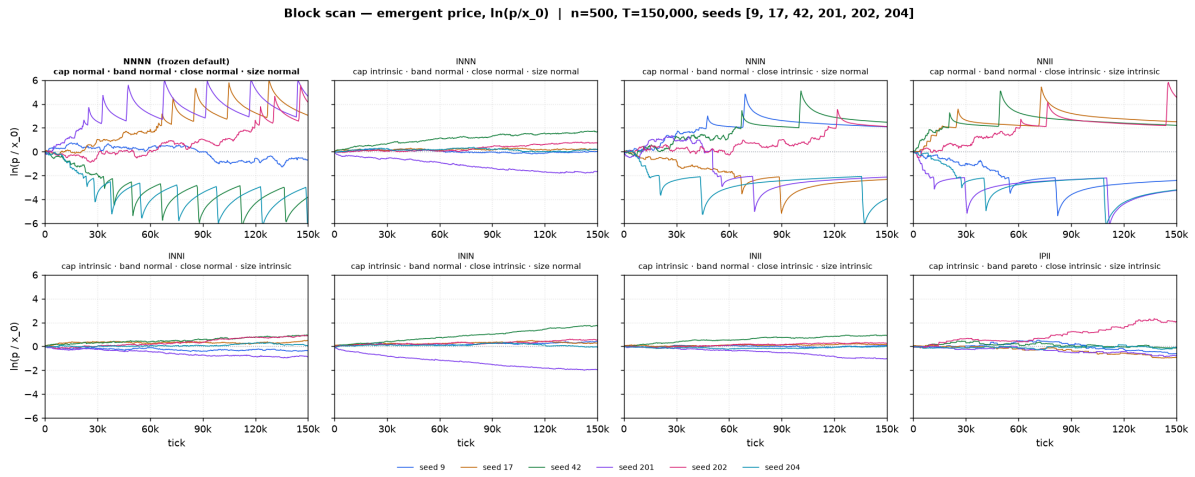


Fig. A17: Introducing intrinsic time. One dial at a time, drawn for each agent, at $n = 500$ for six seeds. The four letters name the dials (capital, bands, closing, and size) with I marking intrinsic time. N refers to the normal distributions of the null model. With the entry clock in intrinsic time (INNN, INNI, ININ, INII) the market stays bounded, whatever else is switched. For NNIN and NNII the saw-tooth is still present. Note that NNNI does not exist (see main text). INII has intrinsic time on the entry clock, the time-out, and the size dial, while keeping the exit bands normally distributed ($1\% \pm 0.3\%$). The last panel is IPII, the same three dials set to intrinsic time, however with heavy-tailed (Pareto) bands, where the threshold is the agent's band.

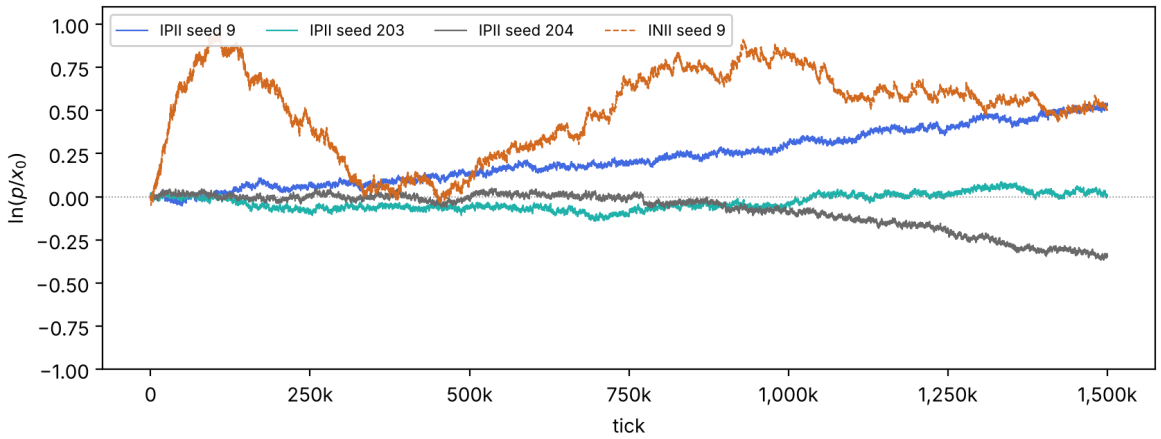


Fig. A18: Price evolutions in intrinsic time. The log price $\ln(p/x_0)$ over 1.5 million ticks, $n = 7{,}500$ per side. The three IPII seeds (solid) wander diffusively. The INII control (dashed/orange) is more volatile as its small-market calm did not survive. IPII trades more and moves less. Compare to Figs. A5 and A6.

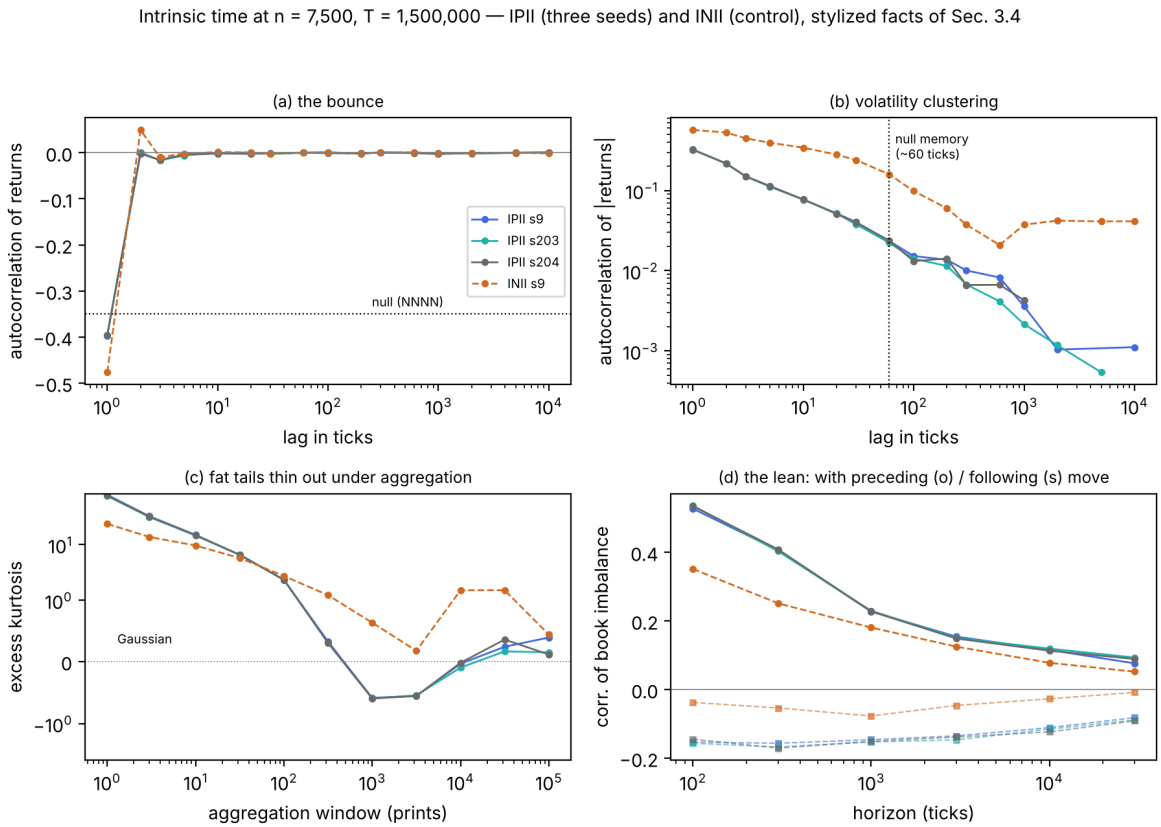


Fig. A19: The stylized facts in intrinsic time. Three seeds of IPII (solid) and the INII control (dashed/orange), $n = 7{,}500$ per side, $T = 1{,}500{,}000$ ticks. (a) The bounce survives: the tick-scale return autocorrelation sits near the null's -0.35 for all arms. Specifically, -0.40 for IPII and -0.48 for INII. (b) Volatility clustering separates the arms: in tick time, IPII's $|r|$-memory is shorter than the null model's ($\sim$20–30 ticks against $\sim$60), and the same holds in event time. INII's stretches to 200–300 ticks—the relaxation time of its entry avalanches. (c) Fat tails thin out under aggregation. IPII reproduces the null model's curve in full—$\sim$80 per print, $\sim$2 over a hundred prints, and close to Gaussian beyond—identically across seeds. INII starts lower per print ($\sim$24) but thins out more slowly. In tick time, IPII's excess kurtosis is 2.7 vs. the null model's 1.7–2.3: the same mild tick-scale tails. (d) The bow wave is intact: the book leans with the preceding move (circles) and predicts the following one (squares) out to 10^4 ticks. The three IPII seeds are indistinguishable at this resolution.

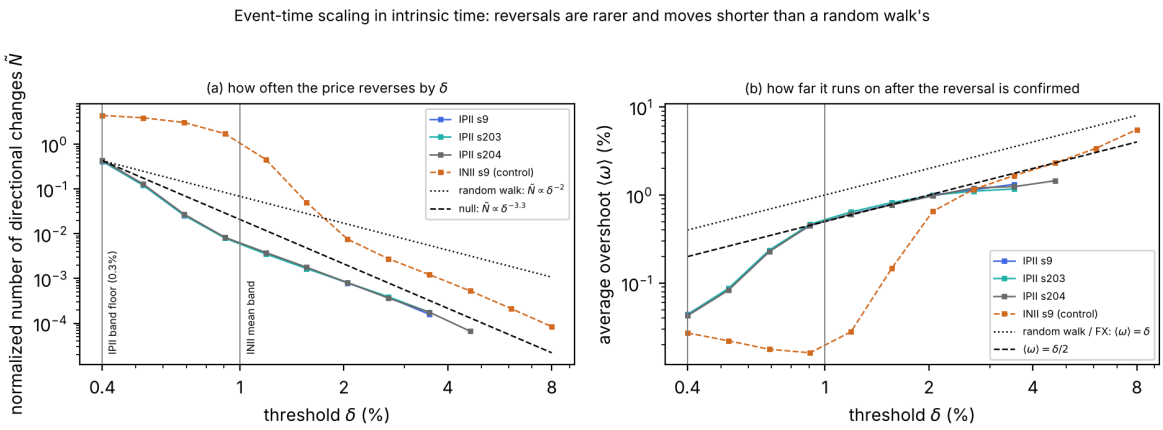


Fig. A20: Event-time scaling in intrinsic time. As in Fig. A11, over thresholds $\delta = 0.4$–8%. (a) The count law: IPII's counts follow $\tilde{N} \propto \delta^{-2.7\ldots-3.0}$ between $\sim$0.9% and $\sim$3.5%—closer to the random walk's δ^{-2} than the null's $\delta^{-3.3}$, and reaching below the single band scale that capped the null's law. INII's partial scaling law is shorter: a plateau below its 1% band, a cliff across 1–2% (local slopes of -7 to -8), and the null's $\delta^{-3.0\ldots-3.4}$ again above 2%. (b) The overshoot: IPII has $\langle\omega\rangle = \delta/2$ from $\sim$0.7% out past 4%, the null model's above-band value extended across the band support. INII again has a shorter interval where scaling is observed. The knee of IPII's laws moves down with its smallest bands; INII's sits above its band, at $\sim$2%, because synchronized exits chain into collective moves larger than any single agent's scale—the band sets the knee only when the crowd does not act as one.

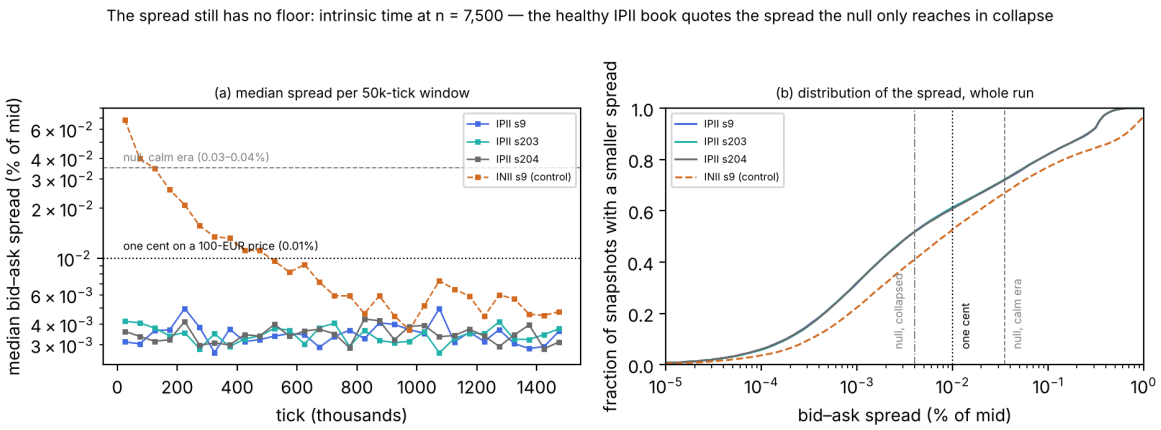


Fig. A21: The spread from intrinsic time. Fig. A13 of the null model reproduced for I_II. Both intrinsic time arms are two-sided within 1% of the mid in effectively every snapshot (IPII 100.0% across three seeds, INII 99.9%). (a) The median spread per 50k-tick window: IPII is stationary from the first window at 0.003–0.005%. This is the null model's collapsed-state spread of 0.004% quoted here by a healthy book. The reason is that the Pareto crowd's smallest bands pave the mid three to ten times more finely. INII never reaches a steady state: its spread decays monotonically for the entire run, from the null model's calm-era value through the one-cent line at ∼500k ticks—the single-scale market's slow transient, in book form. (b) The full distribution: 61% of IPII snapshots quote below one cent on a 100-EUR price; in the null model's calm era, a third of snapshots do.

B. Percentage Calculations

Let $1 > d > 0$ be a fraction corresponding to a percentage and define its up and down reach in log space as

$$d^{\uparrow} = \ln(1+d) > 0, \tag{11}$$
$$d^{\downarrow} = \ln(1-d) < 0. \tag{12}$$

Note that $|d^{\downarrow}| > d^{\uparrow}$. The corresponding percentage moves in log space from a value x_0 are thus $\ln x_0 + d^{\uparrow,\downarrow}$. A rise by d is

$$x_1^{\uparrow} = \exp\big(\ln x_0 + d^{\uparrow}\big) = x_0(1+d), \tag{13}$$

and a fall by d is correspondingly

$$x_1^{\downarrow} = \exp\big(\ln x_0 + d^{\downarrow}\big) = x_0(1-d). \tag{14}$$

As a result, a rise of d followed by a fall of d is then

$$x_2^{\downarrow} = \exp\big(\ln x_1^{\uparrow} + d^{\downarrow}\big) = \exp\big(\ln x_0 + d^{\uparrow} + d^{\downarrow}\big) = x_0(1+d)(1-d) = x_0(1-d^2). \tag{15}$$

As mentioned, this round trip is asymmetric, as $x_2^{\downarrow} \neq x_0$. The inverse round trip follows the same arithmetic.

Note that choosing a single log distance in both directions restores round-trip symmetry. In other words, mixing $d^{\uparrow}$ and $d^{\downarrow}$ in Eq. (15) is the source of the asymmetry, whereas using only $d^{\uparrow}$ or only $d^{\downarrow}$ yields

$$\tilde{x}_2^{\uparrow} = \exp\big(\ln x_0 + d^{\uparrow} - d^{\uparrow}\big) = x_0, \tag{16}$$
$$\tilde{x}_2^{\downarrow} = \exp\big(\ln x_0 + d^{\downarrow} - d^{\downarrow}\big) = x_0. \tag{17}$$

However, restoring symmetry in this way introduces more ambiguity, as two new log thresholds are defined, $-d^{\downarrow}$ and $-d^{\uparrow}$. This means that two new percentage values emerge, next to $x_1^{\uparrow}$ and $x_1^{\downarrow}$, computed as

$$\bar{x}_1^{\uparrow} = \exp\big(\ln x_0 - d^{\downarrow}\big) > x_1^{\uparrow}, \tag{18}$$
$$\bar{x}_1^{\downarrow} = \exp\big(\ln x_0 - d^{\uparrow}\big) > x_1^{\downarrow}. \tag{19}$$

The question now becomes which symmetry convention should be utilized for round trips: $(\pm d^{\uparrow}, \mp d^{\uparrow})$ or $(\mp d^{\downarrow}, \pm d^{\downarrow})$.

A natural choice to avoid the ambiguity is to take the geometric mean of the two arithmetic up-targets $x_1^{\uparrow}$ and $\bar{x}_1^{\uparrow}$ (i.e., combine the log distances $d^{\uparrow}$ and $-d^{\downarrow}$). In detail

$$\sqrt{\exp(\ln x_0 + d^{\uparrow})\,\exp(\ln x_0 - d^{\downarrow})} = \sqrt{\exp\big(2\ln x_0 + d^{\uparrow} - d^{\downarrow}\big)} = \exp\Big(\ln x_0 + \tfrac{1}{2}(d^{\uparrow} - d^{\downarrow})\Big). \tag{20}$$

In other words, we can identify a new log distance defined as the mean of the two old distances as

$$\hat{\delta} = \tfrac{1}{2}\left(d^{\uparrow} - d^{\downarrow}\right) = \tfrac{1}{2}\ln\frac{1+d}{1-d}. \tag{21}$$

Note that the equivalent derivation utilizing the down-targets yields $-\hat{\delta}$.

From Eq. (21), we can define two new percentage values, replacing $x_1^{\uparrow,\downarrow}$ and $\bar{x}_1^{\uparrow,\downarrow}$, guaranteeing symmetric round trips. Specifically

$$\hat{x}_1^{\uparrow} = x_0 e^{\hat{\delta}} = x_0\sqrt{\frac{1+d}{1-d}}, \tag{22}$$

$$\hat{x}_1^{\downarrow} = x_0 e^{-\hat{\delta}} = x_0\sqrt{\frac{1-d}{1+d}}. \tag{23}$$

It follows that

$$\hat{x}_2^{\uparrow} = \hat{x}_1^{\downarrow} e^{\hat{\delta}} = x_0 e^{-\hat{\delta}} e^{\hat{\delta}} = x_0. \tag{24}$$

Again, the inverse round trip follows the same arithmetic.

Note that there is a straightforward simplification of Eqs. (22) and (23) to be found. The justification lies in observing that

$$\hat{\delta} \overset{a}{=} \operatorname{artanh}(d) \overset{b}{=} d + \frac{d^3}{3} + \mathcal{O}(d^5). \tag{25}$$

The equality (a) follows from the fact that inverse hyperbolic functions can be expressed in terms of logarithms [75], and equality (b) is the Taylor series expansion. The difference between $\hat{\delta}$ and d is approximately $d^3/3$ (e.g., 3×10^{-7} for $d = 0.01$). As a result, switching $x \to y$ for notational clarity, d can be substituted for $\hat{\delta}$, yielding

$$y_1^{\uparrow,\downarrow} = y_0 e^{\pm d}. \tag{26}$$

In summary, this is a simple and consistent redefinition of percentage calculations, removing the inherent ambiguities while guaranteeing symmetric round trips. A 10% move reads as $+10.5171\%$ up and -9.5162% down in ordinary percentages, meaning that the normal percentage intuition is given up in exchange for symmetrization.

A note on applicability. The three up-targets agree to leading order by observing their Taylor expansions

$$x_1^{\uparrow}/x_0 = 1 + d, \tag{27}$$

$$\hat{x}_1^{\uparrow}/x_0 = 1 + d + d^2/2 + d^3/2 + \mathcal{O}(d^4), \tag{28}$$

$$y_1^{\uparrow}/x_0 = 1 + d + d^2/2 + d^3/6 + \mathcal{O}(d^4). \tag{29}$$

So both log conventions exceed the arithmetic target by $d^2/2$ and differ from each other only at $\mathcal{O}(d^3)$. For $d = 0.1$ the three up-targets are 1.100, 1.105 and 1.106 times x_0,

a spread of half a percent. At $d = 0.5$ they read 1.50, 1.65 and 1.73. However, at $d = 0.9$ the arithmetic target is $1.9x_0$ while the log conventions give $2.46x_0$ and $4.36x_0$. The symmetrization is therefore a small-move device, appropriate for percentages below roughly 50%, where the shortcut d for $\hat{\delta}$ diverges by at most 10%.

In closing, relating to the existing literature, [76] shows that the log change $\lambda = \ln(x_1/x_0)$ is the only symmetric, additive, and normed indicator of relative change, and proposes using 100λ to calculate percentages, replacing the ordinary notion. In their framework, an up move and its reversal are $\pm\lambda$ by construction, so no two competing log distances arise. In other words, the authors discard the arithmetic percentage d altogether, whereas here the pair $d^{\uparrow} \neq -d^{\downarrow}$ results from retaining it, as seen in Eqs. (11) and (12). Furthermore, Eq. (26) coincides with their proposal for $\lambda = d$, while keeping the ordinary percentage as the parameter. It differs from $\hat{\delta} = \operatorname{artanh}(d)$ of Eq. (25) by $d^3/3$, the threshold required if both log-space definitions derived from d are taken into account and their mean, seen in Eq. (21), is adopted as the natural, consistent choice.

To summarize, for the model, exit bands d from a price x are simply computed as

$$x^{+}_{\text{exit}} = xe^{d}, \tag{30}$$

$$x^{-}_{\text{exit}} = xe^{-d}. \tag{31}$$

Note that for longs, x^{+}_{exit} is the take-profit target, while x^{-}_{exit} is the stop-loss target. For shorts, the opposite holds. Moreover, this choice is further justified, as discussed in Sec 3.3, no result depends on the exact value of d.

C. Scaling Laws

A scaling law is a simple polynomial relationship between a function's input and output. Specifically, a relative change in the input quantity results in a proportional change in the output quantity, independent of the initial size of the input. Scaling laws are scale-invariant, meaning there is no preferred or defining scale, reminiscent of the self-similarity of fractals. They are ubiquitous in complex systems and can be understood as laws of nature of such systems [8, Sec. 6.4].

Scaling-law relations prominently appear in the following domains:

- Allometric scaling laws from biology.
- Scaling-law distributions.
- Scale-free networks.
- Cumulative relations of stochastic processes.

A remarkable allometric scaling law relates the lifespan L and the number of heartbeats H of mammals to their weight M

$$L \approx M^{1/4}, \tag{32}$$

$$H \approx M^{-1/4}. \tag{33}$$

A fundamental invariant of life emerges: the number of heartbeats per lifetime is constant (approximately 1.5×10^9) [77].

Scaling-law distributions have been observed in an extraordinary wide range of natural phenomena: from physics, biology, earth and planetary sciences, economics and finance, computer science, demography to the social sciences [78]:

- The size of cities, earthquakes, moon craters, solar flares, computer files, sand particles, wars deaths, and price moves in financial markets.
- The number of scientific papers written, citations received by publications, hits on web pages, and species in biological taxa.
- The sales of music, books, and other commodities.
- The income of people and the population of cities.
- The frequency of words used in human languages and of occurrences of personal names.
- The areas burnt in forest fires.

The discovery of scale-free networks [25] and their small world properties [79] has been a major breakthrough in network science at the turn of the millennium [80, 81]. Scale-free networks are characterized by a power-law degree distribution, where a few nodes (hubs) have many connections, while most nodes have few. This property has been observed in various real-world networks, including the internet, social networks, and biological systems.

Relating to stochastic processes, the literature goes back to Benoit Mandelbrot [65], the discoverer of the fractal geometry of nature. In finance, particularly in FX markets, a multitude of scaling laws have been described [82, 58, 59]. Two FX scaling laws are of particular interest for this work. The first is the directional-change scaling law, which relating the number of directional changes N in a sample to the directional-change threshold δ [58]:

$$N(\delta) = a\delta^b, \tag{34}$$

where a is a proportionality constant and b is the scaling law exponent. Then, the size of the average overshoot length ω after a directional change is approximately the size of the threshold δ [59]:

$$\langle \omega(\delta) \rangle \approx \delta, \tag{35}$$

where the angle brackets denote the sample average. The directional changes and overshoots are the events defining intrinsic time.

The two FX scaling laws allow for the decomposition of time series into two fundamental components, both only visible through the lens of intrinsic time [73]:

$$\langle r(\Delta t) \rangle_2 \propto \langle \omega(\delta) - \delta \rangle_2 N(\delta), \tag{36}$$

where $r(\Delta t)$ denotes the price changes or returns sampled in physical time, and $\langle x \rangle_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$ is the sample average of the squared values. Note that $\langle \omega(\delta) - \delta \rangle_2$ represents the variability of overshoots (and scales with δ^2). By noting that the overshoots function as a proxy for liquidity [83] and the number of directional changes measures volatility [84], financial time series can be non-trivially decomposed into these two defining characteristics, yielding novel understanding. Moreover, Eq. (36) provides a direct link between physical time and intrinsic time. Finally, the FX scaling laws reveal a numerical relationship between the behavior of a market at small scales to its behavior at larger scales.

D. Attribution and Use of AI

- Parts of this document were compiled from [85] and [86].
- AI tools (Claude and Claude Code from Anthropic, on the Opus 4.8, Opus 5.5, Fable 5 and Fable 5.1 models) were used for drafting, copy-editing, and structural suggestions in Secs. 1 and 2. Secs. 3 and 4 were largely drafted by the AI from the repository record, the author's simulation output, and the author's specifications; the figures, and the analyses behind them, were produced by the AI from tapes the author generated. All AI-drafted passages were reviewed, revised, and endorsed by the author, who takes full responsibility for every claim.
- The code, simulations, and analysis documentation in the repository were written by the AI tools under the author's direction. The author set the research questions, reviewed all results, made every methodological decision, and validated the model manually alongside automated verification. Where the AI proposed an interpretation the author judged unsupported, it was rejected; the retraction ledger in the repository records several such cases.

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